Ministry Approved
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National Textbook Award

Compulsory Education Textbook

Mathematics

SHUXUE

Grade 2 Volume 2

北京师范大学出版社

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BEIJING NORMAL UNIVERSITY PRESS

Dear children,

What are you looking forward to in maths this term?

Wise Old Man

After every maths lesson I like to think about maths questions. This term I hope there will be more interesting ones.

Taoqi

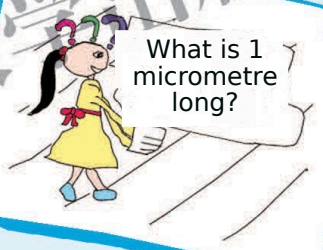
Number Magic

1. Add or subtract 0: no change.

2. Any number × 0 = 0.

3. 0 can make numbers huge.

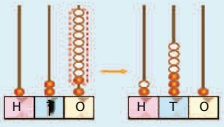
4. What else can 0 do?



What is 1 micrometre long?

Every term I learn new ways to solve maths problems, and I'm sure this term will bring more.

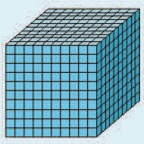
Xiaoxiao



Woof woof! Maths is fun!

Clever Dog

Children, if you want to discover more wonders of maths, let's set off together!



Your friends, the editors

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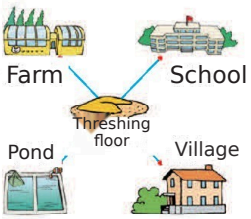

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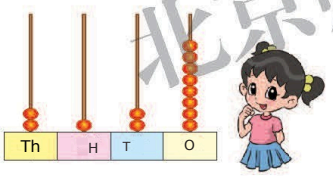

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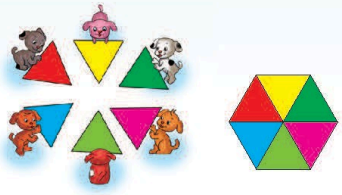

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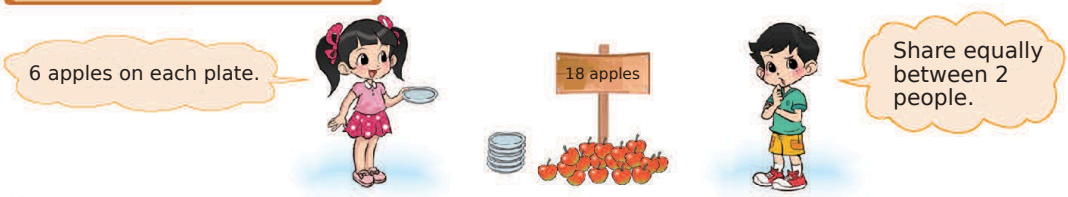
Unit 1 Division

Sharing Apples



- Use the information in the picture to ask a division question.
- With 6 apples on each plate, how many plates do 18 apples fill?

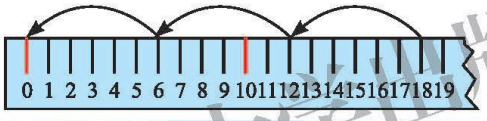
□ ○ □ = □ ( )

1 plate: 6

2 plates: 12

3 plates: 18

18 - 6 - 6 - 6 = 0

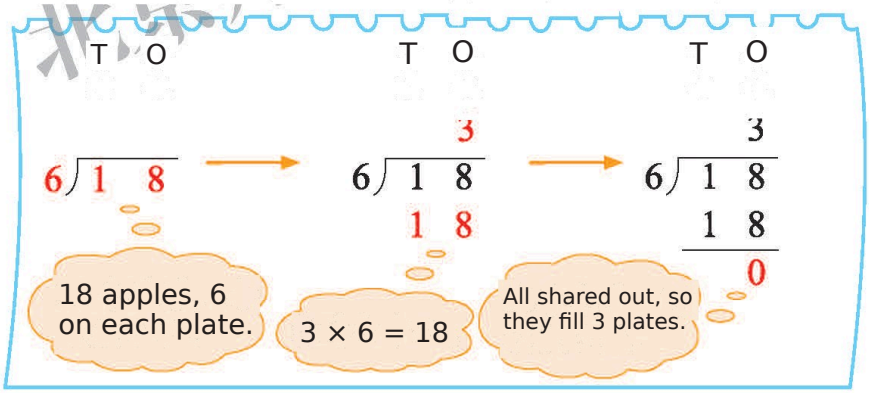


3 × 6 = 18

18 ÷ 6 = 3

Answer: _____。

- We can write these steps as a column division. Look and talk about it.



- Fill in and talk about it.

18 ÷ 2 = □□

63 ÷ 7 = □□

Practice

1. The Lantern Festival

Ask a division question and try to answer it.

2. Circle, fill in, and say what each step of the column division means.

(1) 4 children share 24 crayons equally. Each gets ☐ crayons.

☐ ÷ ☐ = ☐ ( )

☐
☐ ☐ ☐
☐ ☐
☐

(2) 36 balloons, tied in bunches of 6, make ☐ bunches.

☐ ÷ ☐ = ☐ ( )

☐
☐ ☐ ☐
☐ ☐
☐

3.

$4\overline{)8}$	$8\overline{)64}$	$9\overline{)81}$	$6\overline{)48}$
$8\overline{)32}$	$9\overline{)54}$	$7\overline{)42}$	$5\overline{)45}$

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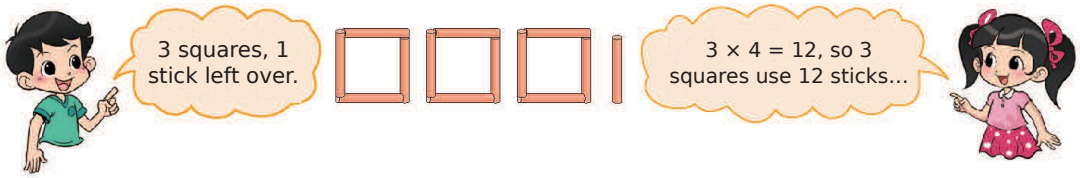
Build (1)

How many squares can 13 sticks make? How many are left over?

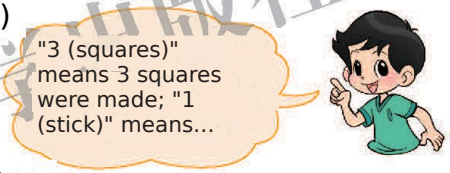


Answer: _____.

Learn the names and think.

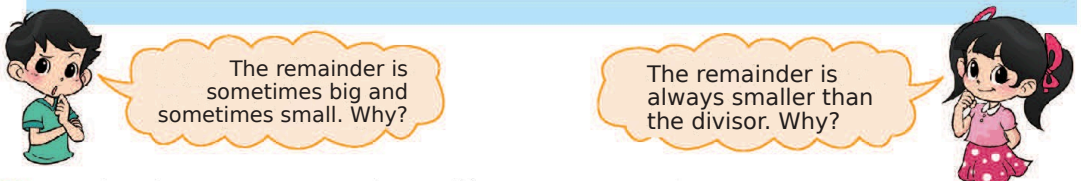
13÷4=3 (squares) ····· 1 (stick)

Remainder



Build and fill in. What do you notice?

Sticks	Squares made	Number sentence
14		14÷4=3 (squares) ····· 2 (sticks)
15		□÷□=□ ( )·····□ ( )
16		□÷□=□ ( )·····□ ( )
17		□÷□=□ ( )·····□ ( )
18		□÷□=□ ( )·····□ ( )
19		□÷□=□ ( )·····□ ( )
20		□÷□=□ ( )·····□ ( )

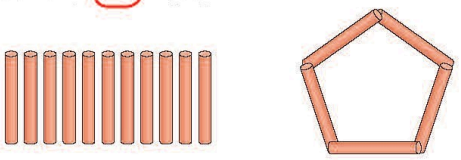


Now try building squares with 25 sticks and with 31 sticks.

Practice

1. Arrange and fill in.

16 sticks. Every 5 sticks make one ▢, making ▢ ▢, with ▢ left over.



▢ ÷ ▢ = ▢ ( ) ····· ▢ ( )

27 apples. Every 6 make one ▢, making ▢ ▢, with ▢ left over.

▢ ÷ ▢ = ▢ ( ) ····· ▢ ( )

2. Build this with sticks. Try it and fill in.

Number of sticks	Shapes made	Number sentence	Number of sticks	Shapes made	Number sentence
14		14 ÷ 7 =	18		
15			19		
16			20		
17			21		

3. Forest Doctor

4 sticks make one square; with 33 sticks, how many squares can you make, and how many sticks are left over?

33÷4= 7 (squares) ..... 5 (sticks)

33÷4=8 (sticks) ..... 1 (stick)

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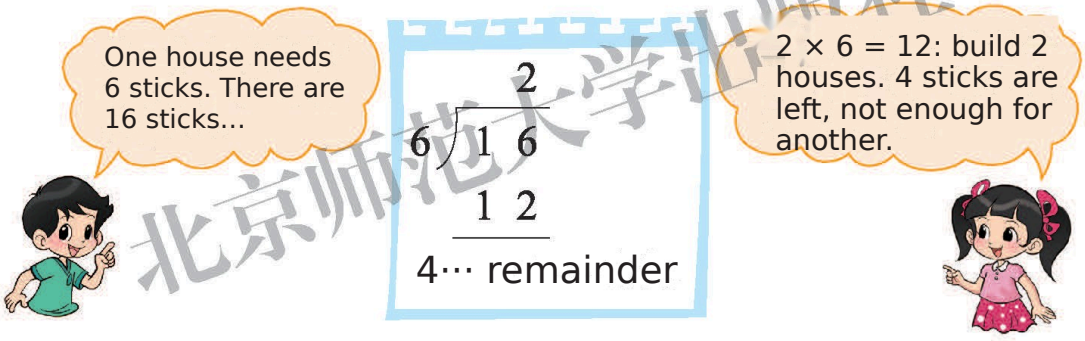
Build (2)

How many houses can you make with 16 sticks, and how many sticks are left over? Build and fill in.

$16 \div 6 =$ ☐ ( ) ····· ☐ ( )

Answer: _____.

Think about building the houses above. Look and say.



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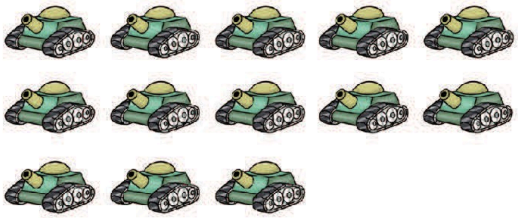
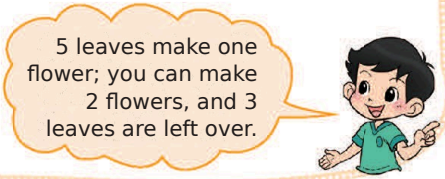
Look and say.

$13 \div 5 = 2 \cdots\cdots 3$

5 children share them equally.



5 leaves make one flower.



Practice

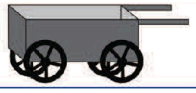
1. (1) Each child gets ☐ blocks, and ☐ are left.

$$3\overline{)20}\quad \begin{array}{r} \square \\ \square\square \\ \hline \square \end{array}$$

(2) Using the sharing, tell a partner what each step of the column method means.

2. How many of each vehicle can 15 wheels make? How many are left?

(1) Fill in.

			
Vehicles made			
Wheels left			

(2) Work out each division using the column method.

3. $4\overline{)12}$ $2\overline{)9}$ $3\overline{)17}$ $5\overline{)21}$

4. ☐ What is the largest number that fits?

$4 \times \square < 17$ $\square \times 5 < 43$ $\square \times 8 < 38$

$6 \times \square < 25$ $\square \times 7 < 62$ $9 \times \square < 57$

5. Think of a problem you can solve with "10 ÷ 4 = 2 ..... 2" and tell your partner about it.

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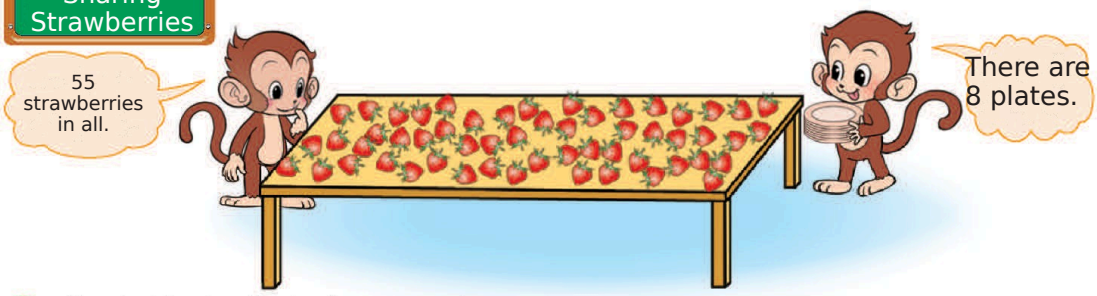
Sharing Strawberries



If we share them equally, how many 🍓 on each plate? How many are left?

$55 \div 8 = \square\ (\ )\cdots\cdots \square\ (\ )$

Is your answer right? Circle the picture to check.

$$\begin{array}{r} \square \\ 8\overline{)\,5\ 5} \\ \square\square \\ \hline \square \end{array}$$

7×8=56, more than 55; 6×8=48, less than 55; so the quotient is...

Answer: _____。

7 🍓 go on 1 plate. How many plates can you fill? How many are left?

$\square\square \div \square = \square\ (\ )\cdots\cdots \square\ (\ )$

What if each plate holds 6?

$$\begin{array}{r} \square \\ \square\overline{)\,5\ 5} \\ \square\square \\ \hline \square \end{array}$$

Answer: _____。

Are the calculations below correct? Think and discuss.

56 is greater than the dividend!

$$\begin{array}{r} 8 \\ 7\overline{)55} \\ 56 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 8 \\ 6\overline{)55} \\ 48 \\ \hline 7 \end{array}$$

The remainder must be ... than the divisor.

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Practice

1. Arrange flowers.

22

Circle to check. Is your answer right?

(1) If there are 3 vases, how many flowers go in each, and how many are left?

(2) If each vase holds 5, how many vases can you fill, and how many are left?

2. ☐ What is the largest number that fits?

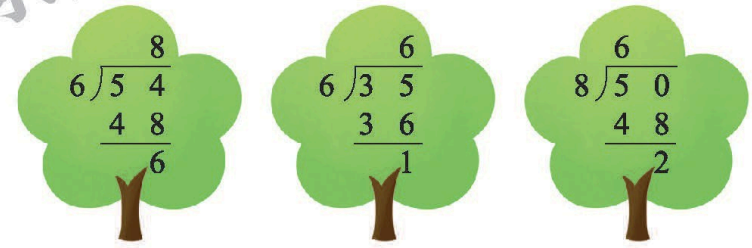
☐ × 6 < 41　　4 × ☐ < 26　　5 × ☐ < 39

☐ × 8 < 73　　☐ × 9 < 60　　7 × ☐ < 31

3. Use the column method.

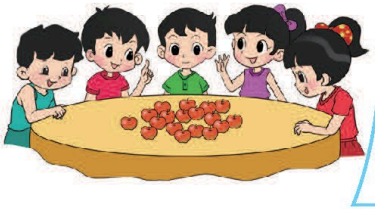
20 ÷ 6 =　　34 ÷ 4 =　　30 ÷ 7 =　　27 ÷ 3 =

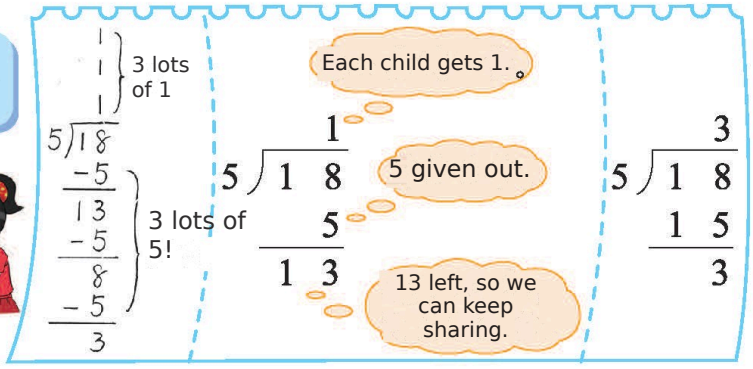
4. Forest Doctor



5. Look, think, and explain what happens as the apples are shared.

There are 18 apples to share equally among 5 children.





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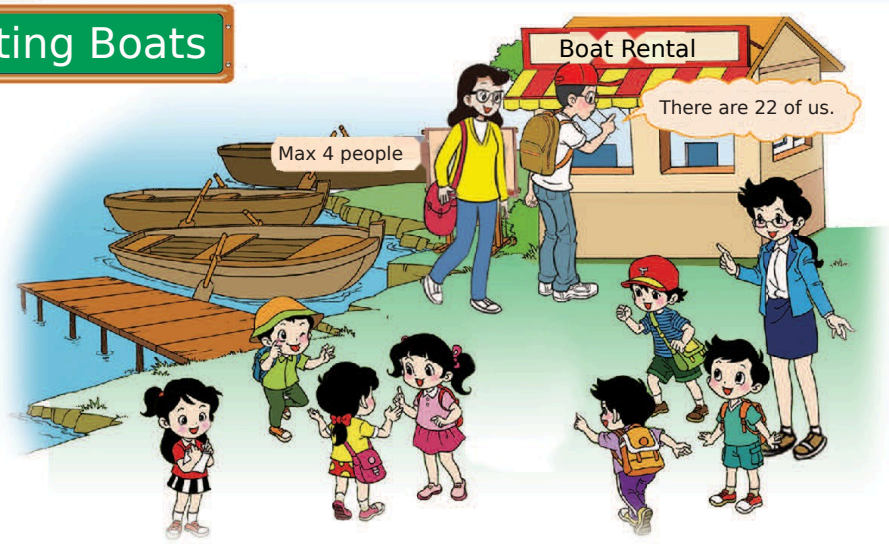
Renting Boats

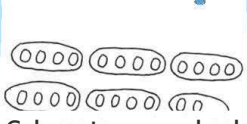


What maths information does the picture show? Discuss.

22 people...

"Max 4 people" means each boat can take 1, 2, 3 or 4 people, but no more than 4.

What is the smallest number of boats needed? Try it and discuss.



6 boats needed

1 boat: 4
2 boats: 8
3 boats: 12
4 boats: 16
5 boats: 20

2 left

6 boats needed

Oh, I see! The 2 people left over need 1 more boat, so at least 6 boats are needed.

Can you solve it with a number sentence?

4 people go in 1 boat, so we can write a division number sentence...

The remainder is 2. These 2 people need 1 more boat.

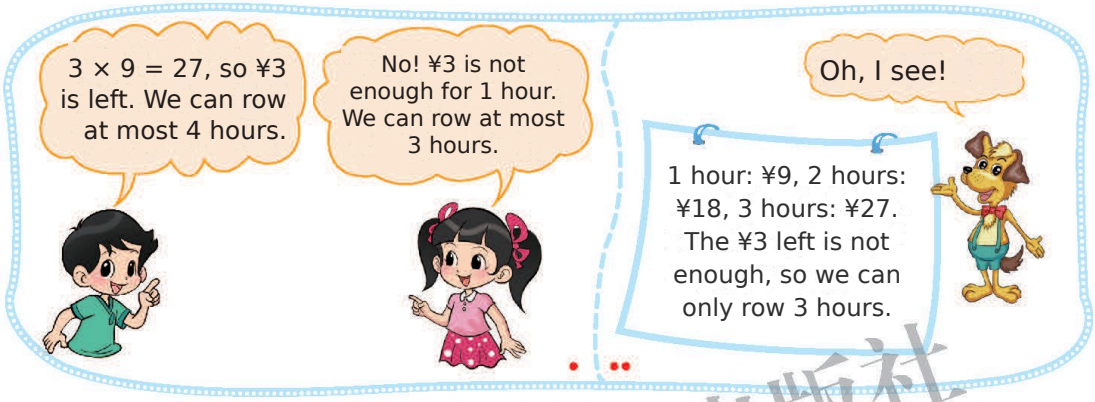
□○□=□( )……□( )

Answer: _____。

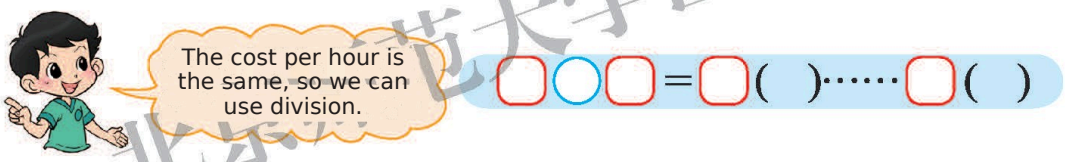
Try It

You have ¥30. If you rent 1 boat, how many hours can you row at most? Think and discuss.

Each boat costs ¥9 an hour.



Can you solve the problem above with a number sentence?



Answer: _____。

Practice

1. (1) What maths information is in the picture? Talk with your partner.

(2) What is the smallest number of tables needed? Draw and calculate.

2. What is the smallest number of leaves 40 ants need? Draw and calculate.

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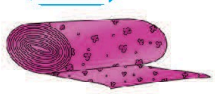
3.

How many outfits can be made at most?

1 outfit takes 3 m of floral cloth.

25 m



How many stars can be folded at most?

1 star takes 9 cm of ribbon.

70 cm

4. If each parent gets 1 cup of water, what is the smallest number of bottles needed?

Welcome to the Parents' Meeting

38 parents have come.
1 bottle fills 9 cups.

5. Use the column method.

$47 \div 9 =$	$32 \div 7 =$	$41 \div 8 =$
$42 \div 6 =$	$27 \div 4 =$	$37 \div 5 =$

6. Going to the airport.

Max 7 passengers

Max 4 passengers

25 people

(1) If all go by minibus, what is the smallest number of minibuses needed?

(2) If all go by car, what is the smallest number of cars needed?

(3) If minibuses and cars can be mixed, give one plan for the trip.

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Exercise 1

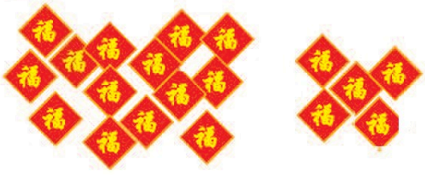
1. Circle and fill in. Use the sharing to explain each step of the column method.

4 children share 21 cards equally. Each gets ☐ cards, and ☐ are left.

☐○☐=☐( )……☐( )

☐
☐ ☐☐
☐☐
☐

19 Fu cards, 5 cards to 1 bag. They fill ☐ bags, and ☐ are left.



☐○☐=☐( )……☐( )

☐
☐ ☐☐
☐☐
☐

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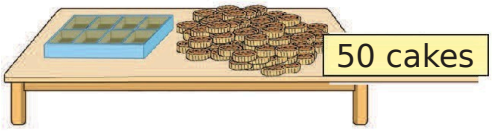
2.



(1) Fill in.

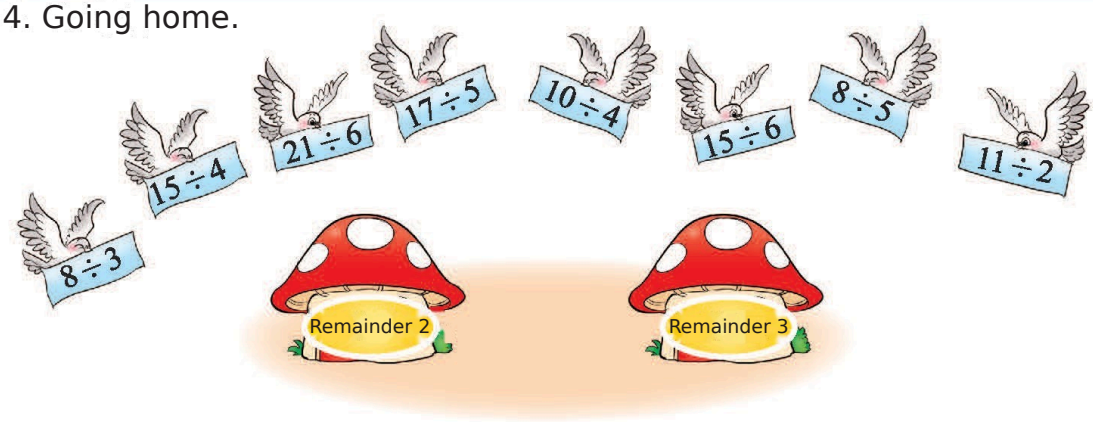
Mooncakes	8	16	24	32	40	48	56
Boxes	1	2					

(2) How many boxes can be filled, and how many mooncakes are left? Write a number sentence and calculate.

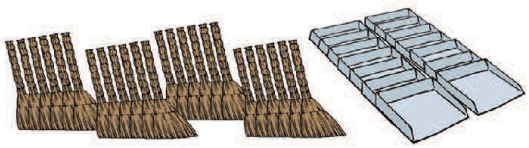
3. Use the column method.

52 ÷ 8 =　　46 ÷ 9 =　　40 ÷ 7 =　　24 ÷ 6 =

4. Going home.



5. Look and fill in.



24 brooms 14 dustpans

Shared equally among 6 classes, each gets

☐ brooms;

☐ dustpans, ☐ left.

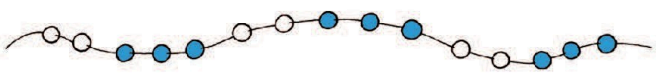
6.

(1) At least how many sleds do 19 snow leopards need?

(2) At most how many jackets can 27 buttons be sewn onto?

7. In this string of beads, what colour is the 18th bead from the left?

What about the 25th? The 34th?



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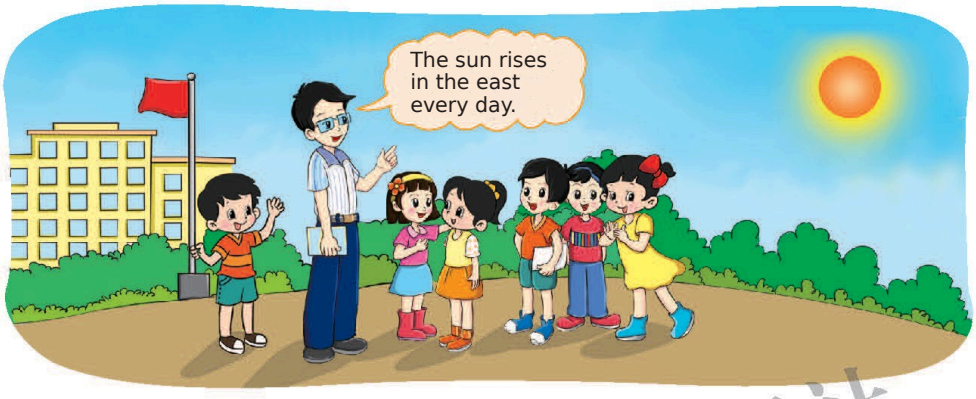
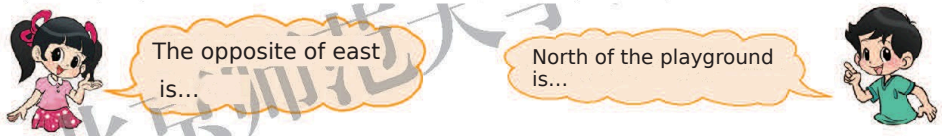
Unit 2 Direction and Position

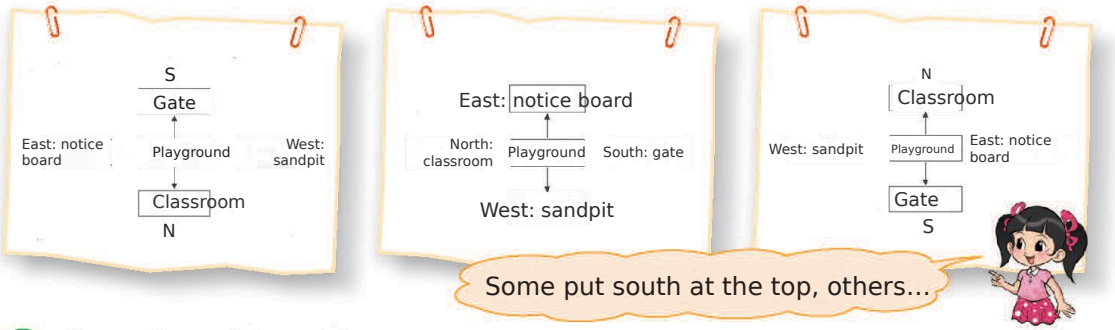
Four Directions



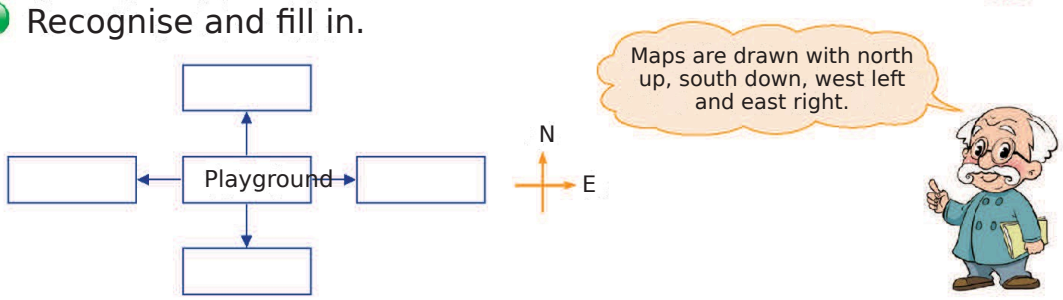
Go to the playground and find east, south, west and north. Say what there is in each direction. Record it in Figure 1 of Appendix 1 and mark the directions.



Back in the classroom, put your record sheets on the blackboard. Look and talk about them.



Recognise and fill in.



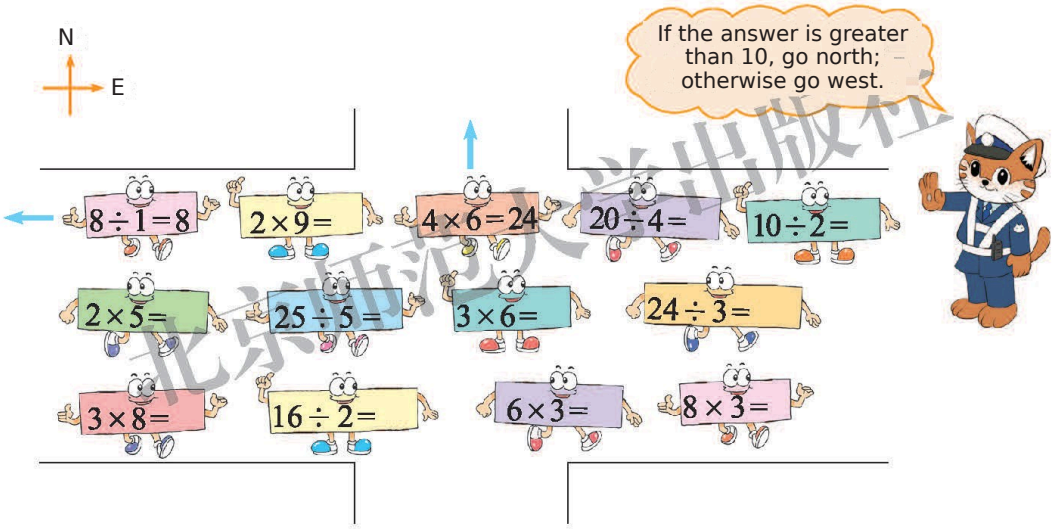
Practice

1. Find east, south, west and north in the classroom.

East: ___, south: ___, west: ___, north: ___.

2. I say, you do.

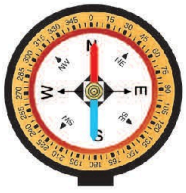
3. Calculate and mark.



4. (1) Talk with Mum and Dad: which direction does your front door face?

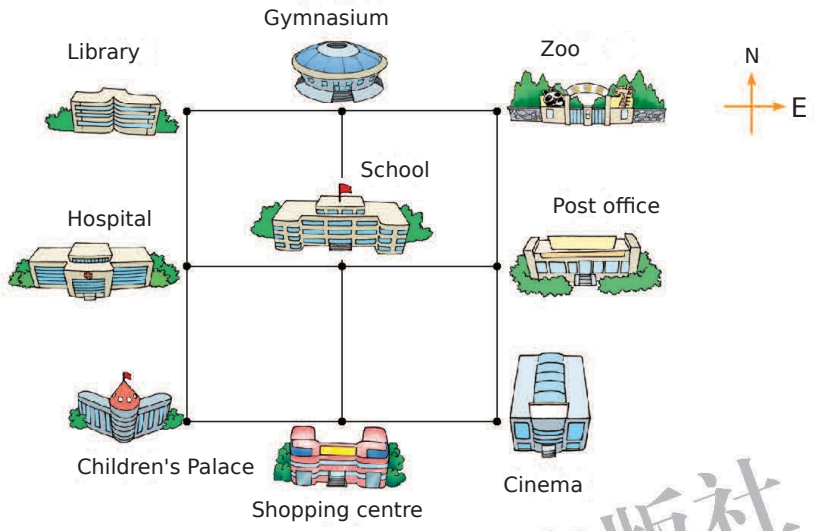
(2) Observe the night sky and find the Big Dipper.

5. Read and tell.



The compass was invented by our ancestors more than two thousand years ago. It is very good at showing directions: wherever it is placed on Earth, one end always points south and the other north. With it, people never lose their way.

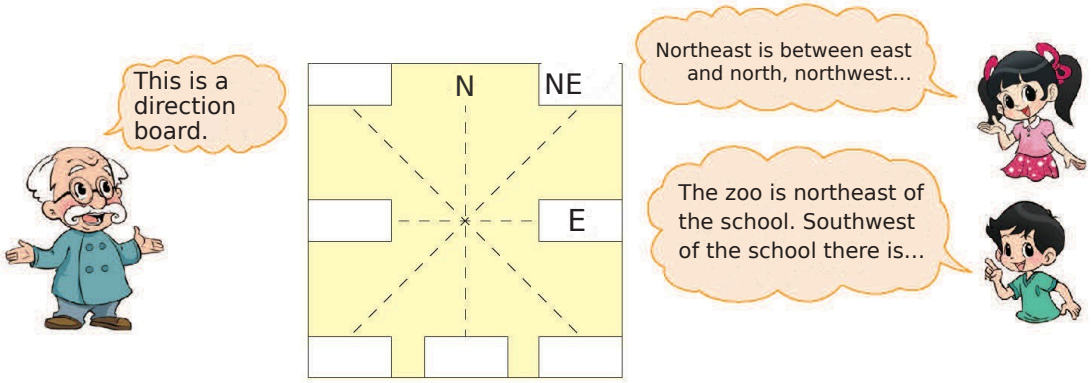
Telling Directions



- Look and fill in.

  The gymnasium is ___ of the school; the shopping centre is ___ of it.
  The hospital is ___ of the school; the post office is ___ of it.

- Identify, fill in and talk.

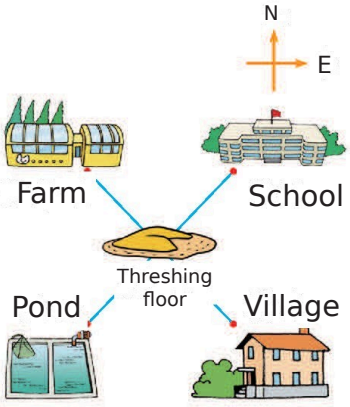


- Make a direction board. Go to the playground, look around, record what is in each direction of the school, then tell your partner.

Practice

1. Use the direction board to fill in the blanks.

(1) The school is ___ of the threshing floor.
(2) The village is ___ of the threshing floor.
(3) The pond is ___ of the threshing floor.
(4) The farm is ___ of the threshing floor.



2. Look at the map and fill in.

South China Sea Islands

Sichuan is roughly ____ of Beijing. Jilin is roughly ____ of Beijing. Taiwan is roughly ____ of Beijing.

3. Calculate and mark.

N
E

35 ÷ 4 =
65 ÷ 9 =
48 ÷ 7=
58 ÷ 7 =
25 ÷ 6 =
46 ÷ 7 =
73 ÷ 8 =
29 ÷ 3 =
43 ÷ 7 = 6……1
53 ÷ 8 =
19 ÷ 5 = 3……4
37 ÷ 4 =
36 ÷ 8 =

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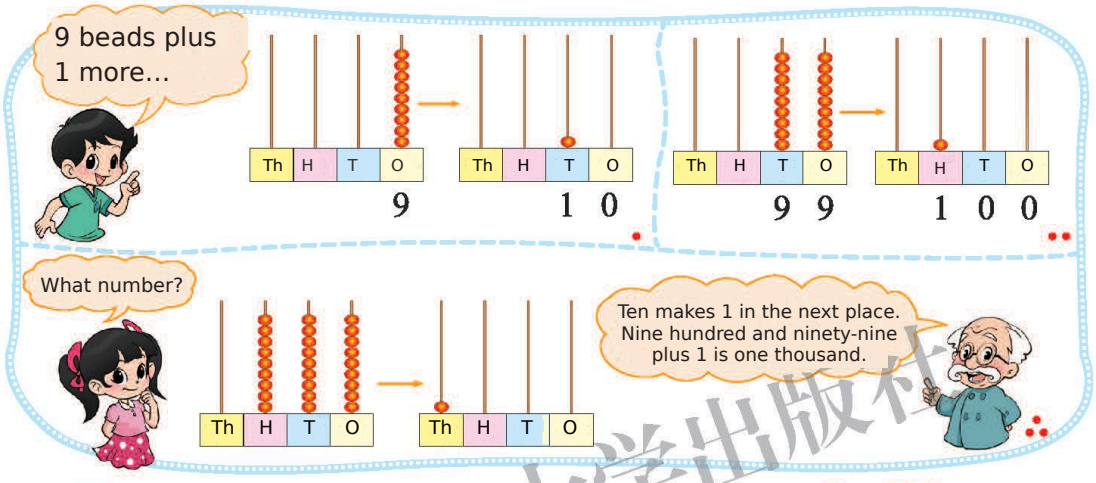
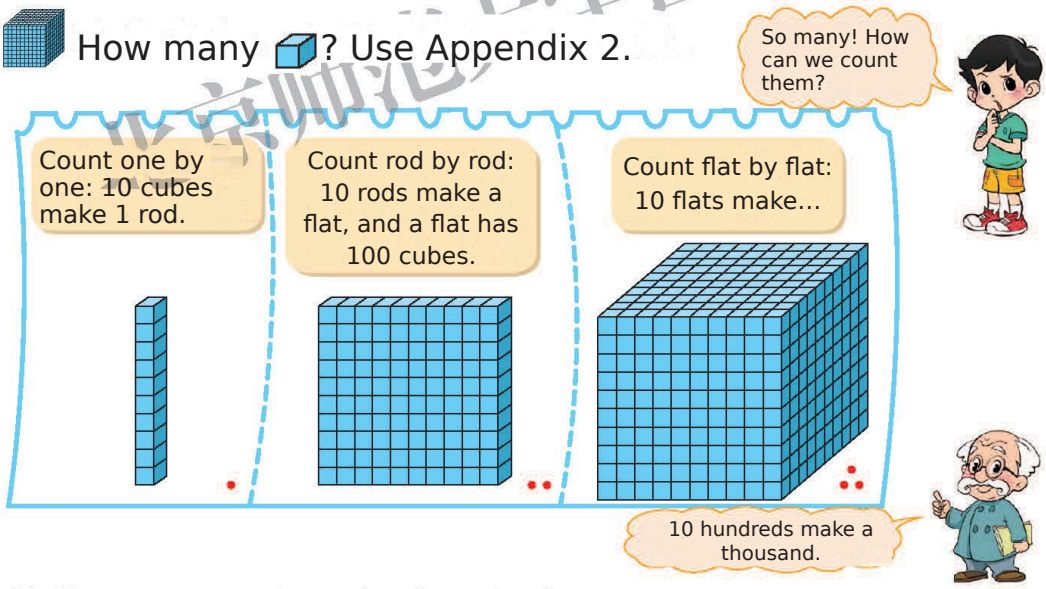
Unit 3 Large Numbers in Life

Count (1)

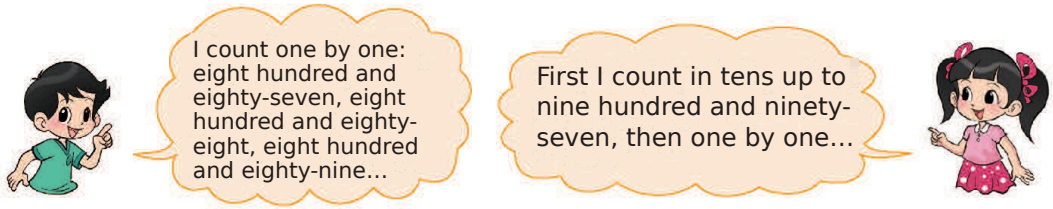
Add 1 more bead. What is it now?



How many ? Use Appendix 2.



Can you count from eight hundred and eighty-seven to one thousand?



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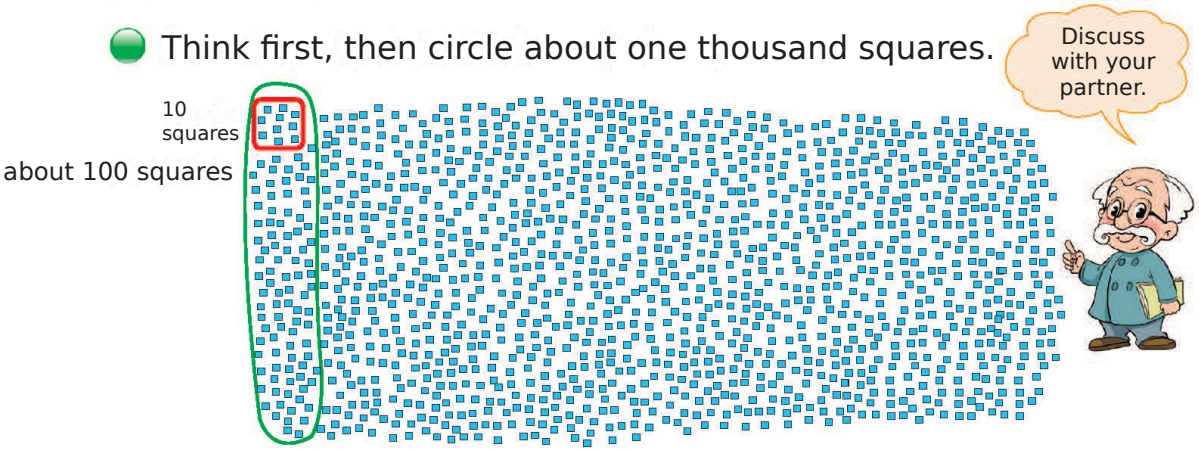
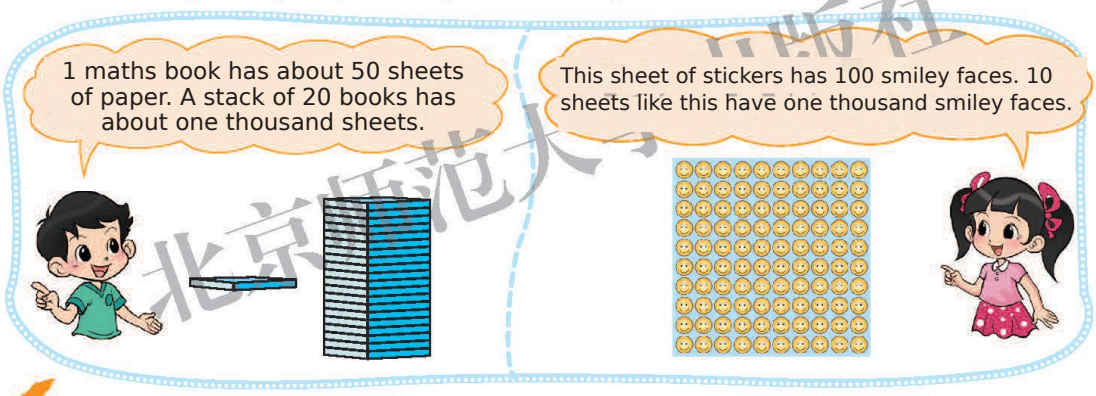
Try It

Think first, then circle about one thousand squares.



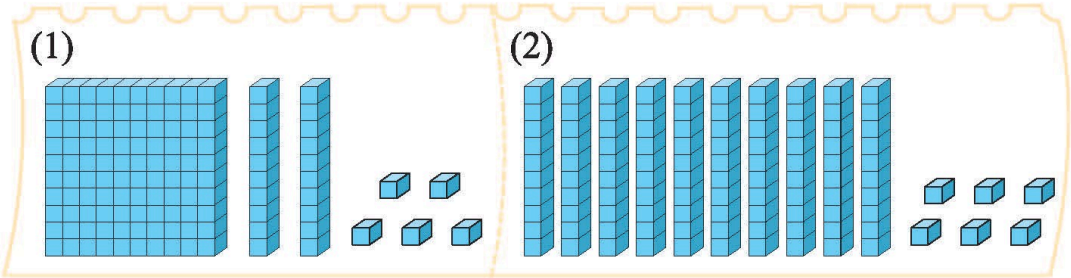
How big is one thousand? Look and talk.



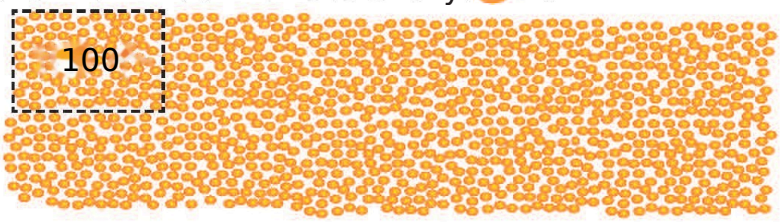
Practice

1. Count one by one from two hundred and ninety-eight to three hundred and two, and show each number on a counting frame.

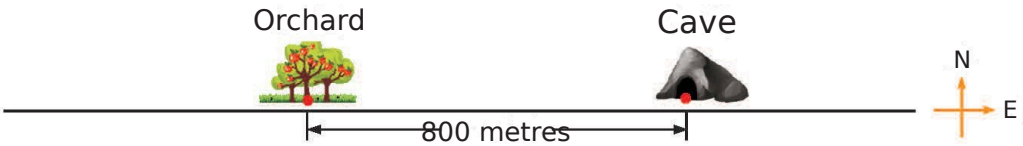
2. Talk about it: how many are there?



3. Estimate and circle. About how many are there?



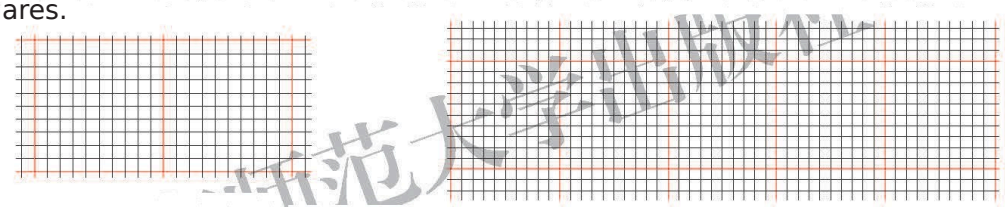
4. Look, then mark the approximate positions of the station and Qingshui Lake. Qingshui Lake is 300 metres east of the cave, and the station is 400 metres west of the orchard.



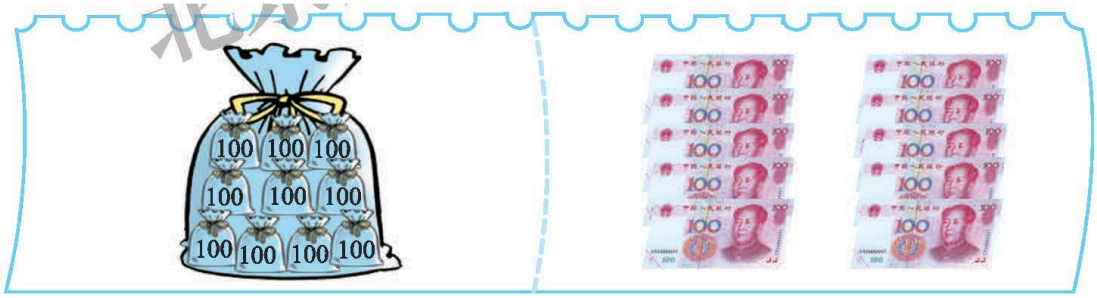
5. Colour in.

(1) Colour one hundred and thirty-eight squares.

(2) Colour three hundred and six squares.



6. Count and talk. How much money altogether?



7. Group game: try it with the pictures in Appendix 2.

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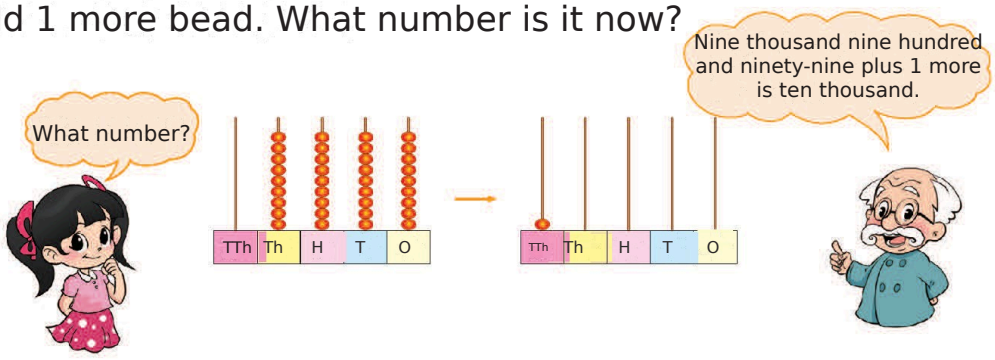
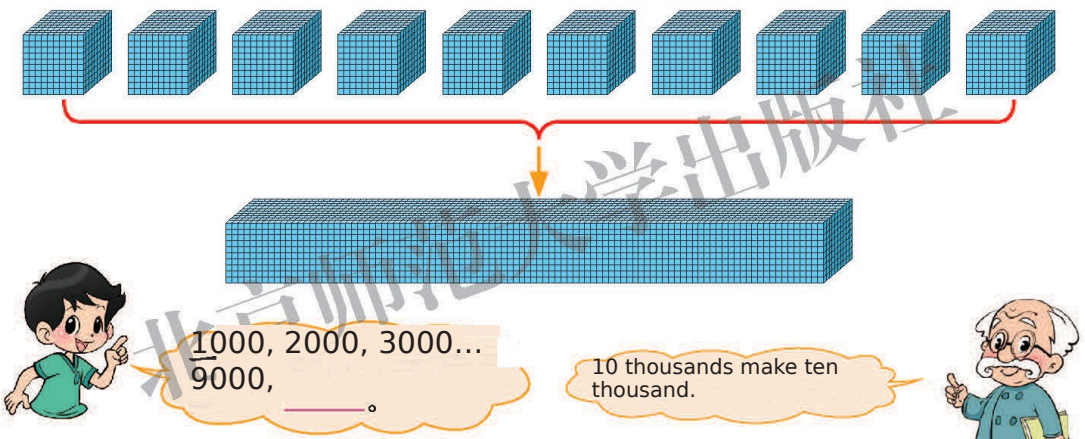
Count (2)

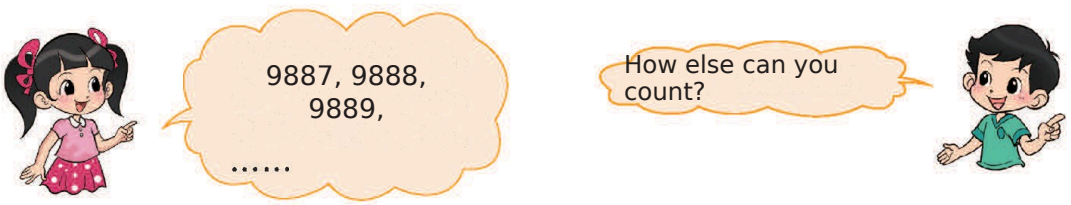
Add 1 more bead. What number is it now?



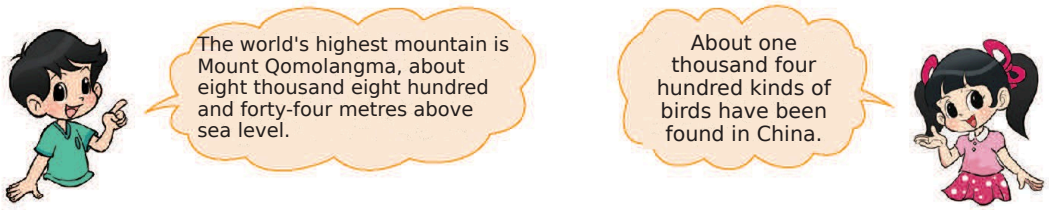
Count.



Can you count from 9887 to 10000?



Talk about large numbers in everyday life.



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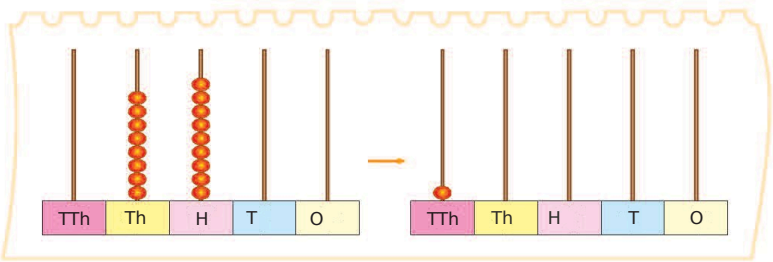
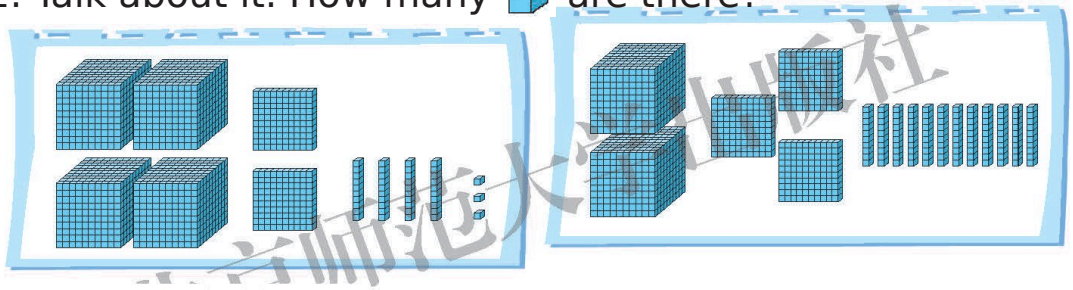
Practice

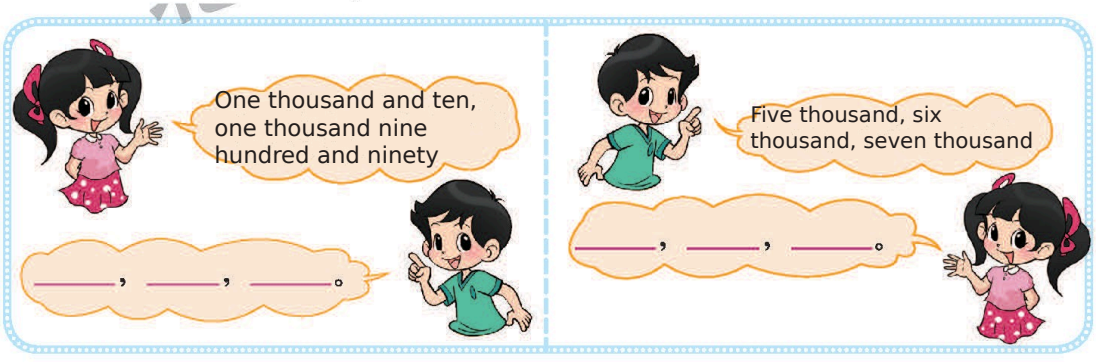
1. Count by hundreds from eight thousand nine hundred to ten thousand. Show each number on a counting frame.



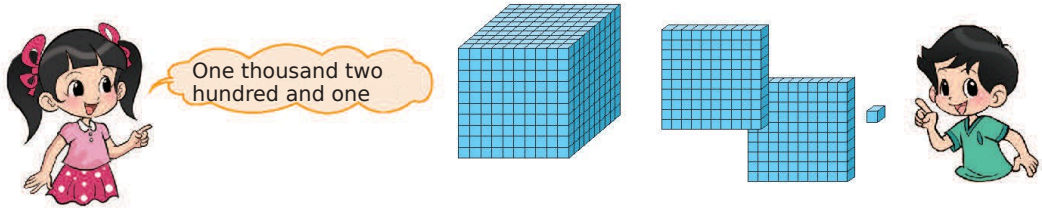
2. Talk about it. How many are there?



3. Think and talk.



4. I say a number and you build it. Use the pictures in Appendix 2.



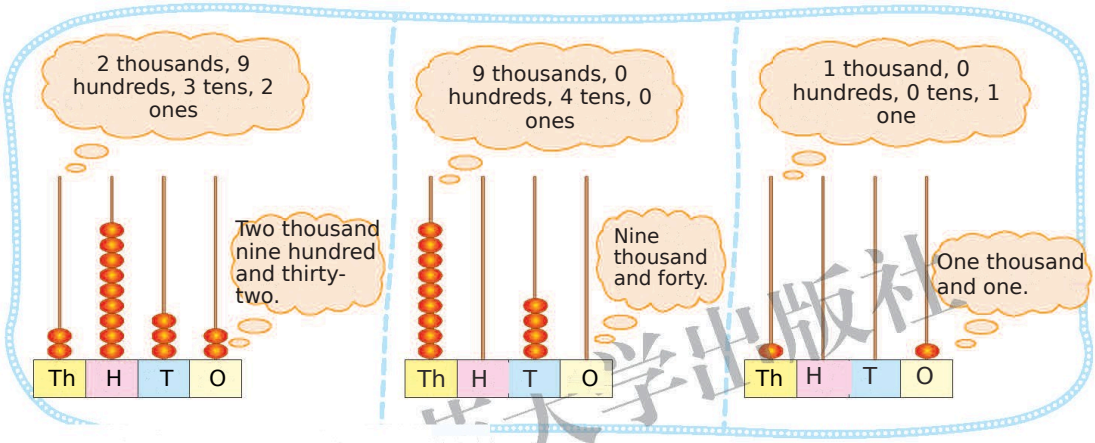
Move the Beads

Fill in and learn the place values.

Place-value chart

.....	( )	( )	( )	Tens	Ones

Move the beads, say the number, and read it!



Look and write.

three hundred and twenty-seven

eight thousand and fifty-two

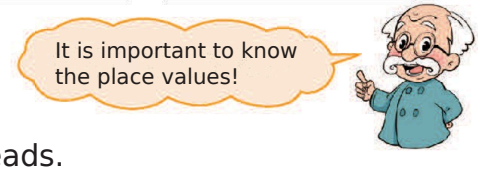
one thousand five hundred and three

Write:

TTh	Th	H	T	O
		3	2	7

TTh	Th	H	T	O

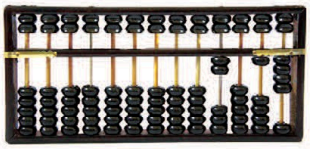
TTh	Th	H	T	O



Get to know the abacus. Move the beads.

The abacus is a calculating tool invented by the Chinese. It has a history of more than one thousand years. To count on an abacus, first decide on the ones place. In the ones place, 1 lower bead stands for one and 1 upper bead stands for five.

When the ones place reaches ten, carry 1 to the place on its left (the tens place).



Talk about it. How did you show 2014 on the abacus?

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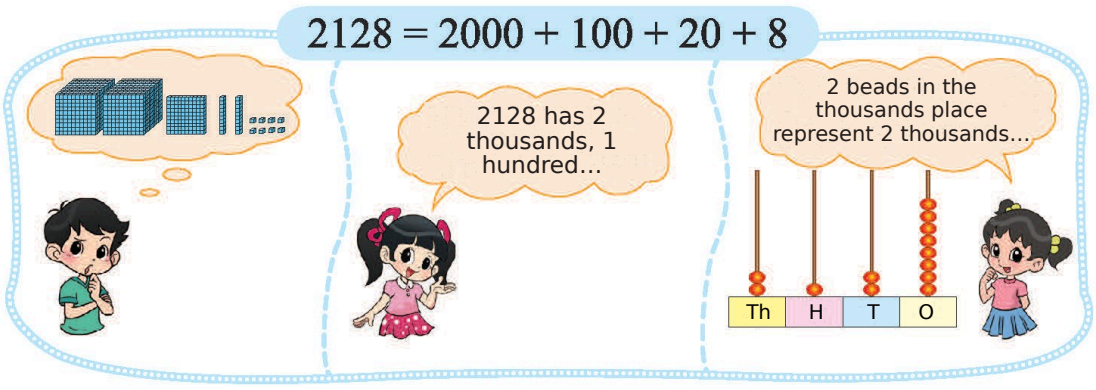
Try It

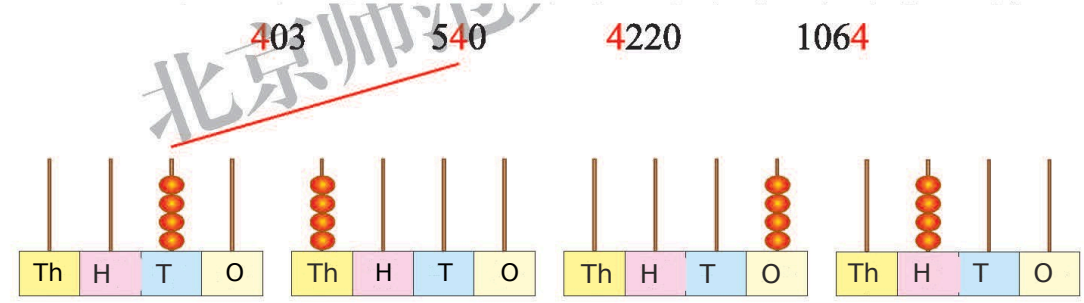
Look and say.



Can you find another number and talk about it?

Practice

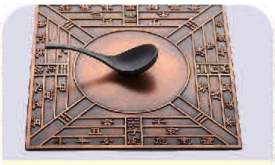
1. Think. What does the "4" in each number stand for? Draw lines to match.



2. Read. Write each number of years in digits and show it on the abacus.

Gunpowder weapons: about one thousand one hundred years ago.

( )



The compass: about two thousand five hundred years ago.

( )

Bi Sheng's movable type: about nine hundred and seventy years ago.

( )

Cai Lun's improved paper: about one thousand nine hundred years ago.

( )

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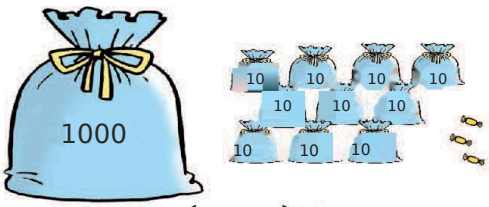
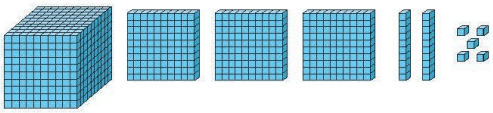
3. Count and fill in.

How many sweets are there?



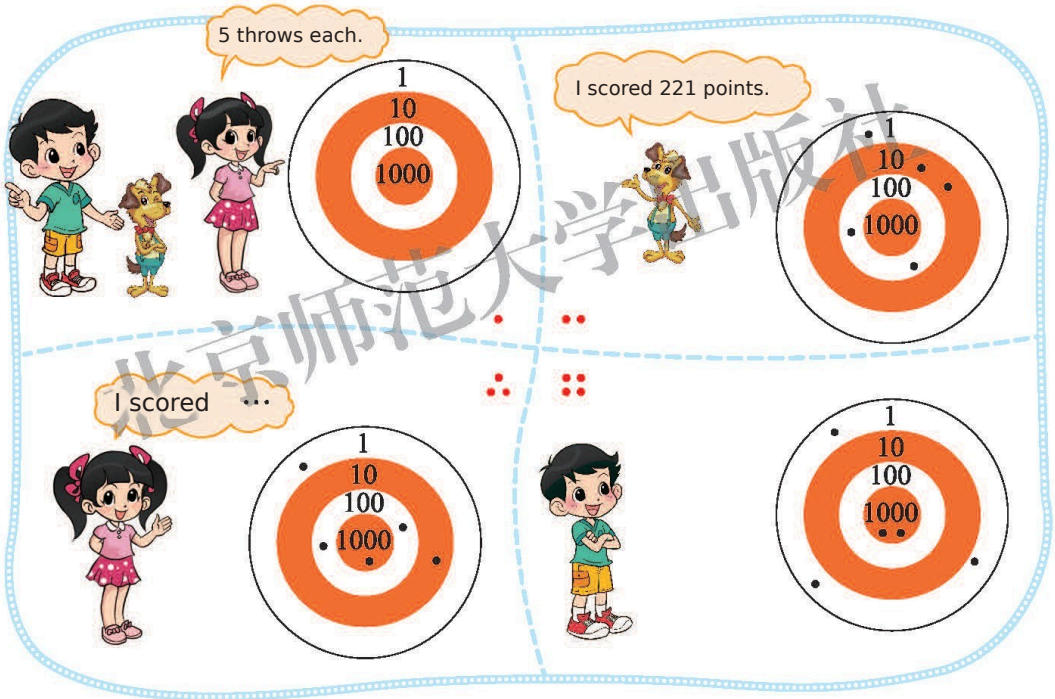
( )

How many are there?



□ = □ + □ + □ + □

4. Maths game: Who comes first?



5. Read and tell.

How the ancient Chinese recorded numbers

The earliest Chinese way of recording numbers appeared in oracle bone inscriptions from the 16th to the 11th century BC. For example, 2656 was written as =∩X∩. For convenience, from the 5th century BC numbers were also shown with counting rods. For example, 6728 was written as ⊥TT=TTT.

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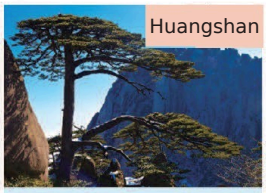
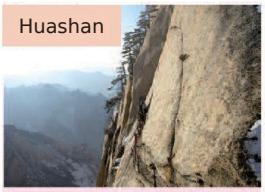
Compare



About 1865 m above sea level

About 575 m above sea level

About 1533 m above sea level



About 2155 m above sea level

Which is the lowest?

TTh	Th	H	T	O	
	1	8	6	5	Huangshan
		5	7	5	Xiangshan
	1	5	3	3	Taishan
	2	1	5	5	Huashan

Xiangshan is lower than 1000 m. The others are all...

4-digit numbers are greater than 3-digit...

Which is the highest?

TTh	Th	H	T	O	
	1	8	6	5	Huangshan
	1	5	3	3	Taishan
	2	1	5	5	Huashan

Huangshan and Taishan are both over 1000 m, and Huashan...

All three are four-digit numbers. First look at the highest place...

Which is higher, Huangshan or Taishan? Which is lower?

The digit in the hundreds place of 1865 is 8, which represents 8 hundreds. Taishan...

TTh	Th	H	T	O	
	1	8	6	5	Huangshan
	1	5	3	3	Taishan

Fill in from smallest to largest.

( )<( )<( )<( )

Say how you compared them.

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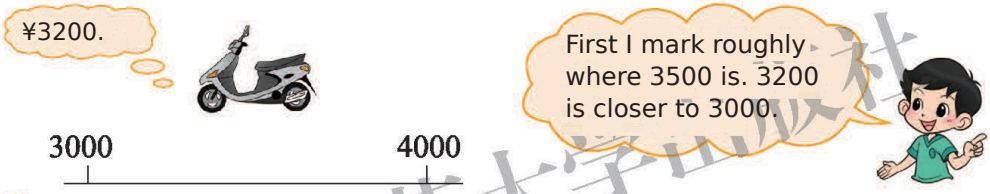
Try It

Which bike is the cheapest? Mark and compare.

①	②	③
¥1200	¥800	¥1500

( ) > ( ) > ( )

Mark the approximate position of 3200 on the number line.



Practice

1. Harvesting Corn

(1) ( ) harvested the most corn and ( ) harvested the least. Say how you worked it out.

(2) Arrange the amounts of corn harvested by the four animals in order from largest to smallest.

( ) > ( ) > ( ) > ( )

2. Fill ◯ with > or <. Say how you compared them.

1074 ◯ 846　　389 ◯ 982　　5102 ◯ 5184

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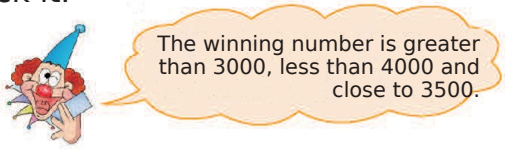
3. Fill in.

4. The highest mountain in Taiwan, China, is Yushan, with an altitude of about 3952 m. Mark the approximate position of 3952 on the number line.

5. There is a prize draw before the circus show. Which is the winning number? Tick it.



☐ 2980 ☐ 4001
☐ 3408 ☐ 3920

6. Maths game.

(1) Work in pairs: one person thinks of a number and the other guesses it.

(2) The guesser asks questions and may only be told "yes" or "no".

Keep guessing, and use as few guesses as possible.

How Many Words?

Zu Chongzhi, an ancient Chinese mathematician, loved reading from childhood, and people praised him as a learned young man. He often observed the movements of the sun and other heavenly bodies. His figure for the number of days the moon takes to go around the earth was very precise, differing from modern scientific results by less than 1 second.

Zu Chongzhi is respected all over the world. In Paris, France, the "Palace of Discovery" science museum shows Zu Chongzhi's name and the value of pi he found. Western scientists even named a crater on the moon "Zu Chongzhi Crater".

Estimate: about how many words are in this story?

Work in groups, estimate and discuss.

Estimate and fill in

About 100 grains.

About ( ) grains.

About ( ) grains.

Practice

1. Estimate and discuss: about how many words are there? (30, 300 or 3000 words)

Dinosaurs first appeared about 235 million years ago and died out about 65 million years ago in the Cretaceous mass extinction.

In fact, people have been finding dinosaur fossils for a long time. Long before dinosaurs were identified, Europeans knew of many huge, strangely shaped bone fossils underground. Only after dinosaurs were found and compared with lizards did scientists decide they were long-extinct, lizard-like reptiles.

In 1842, the British paleontologist Richard Owen created the English word "dinosaur". At that time, few dinosaurs had been found. Since dinosaurs were found in Antarctica in 1989, dinosaur remains have been found on all seven continents. Today, at least 650–800 kinds of dinosaurs (paleontological species) have been described. Later, scholars in China, Japan and other countries translated the word as "terrible dragon", because these countries have long had dragon legends and see the dragon as the king of scaly creatures; the snake, for example, has long been called a "little dragon".

2. Compare and estimate.

3. Estimate: how many jelly beans are there?

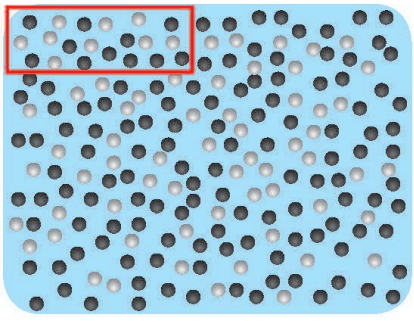
There are 50 beans.

About ( ) beans.

About ( ) beans.

4. About how many stones are there?

Estimate and talk about it.



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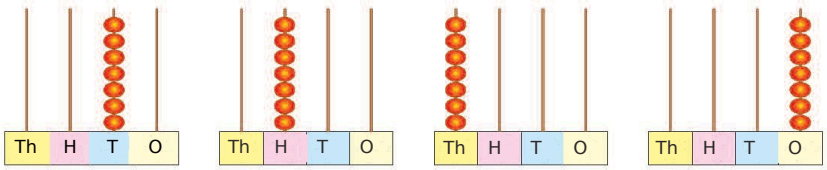
Exercise 2

1. Read, match, and say what the "7" in each number represents.

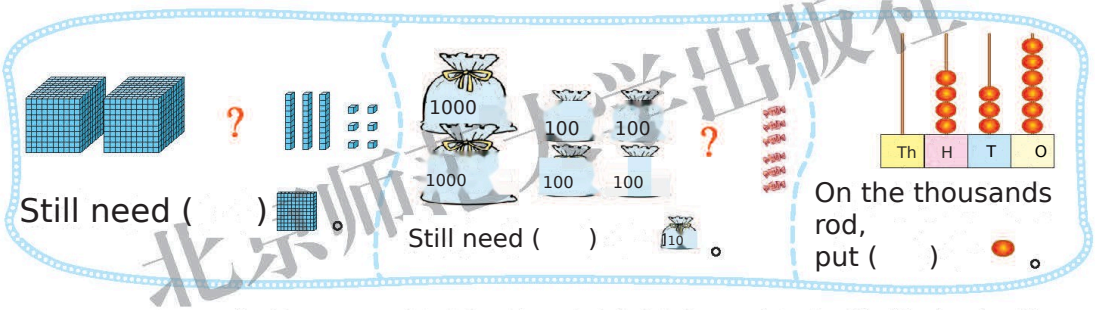
307　　754　　371　　7593



2. In ☐ write a suitable digit.

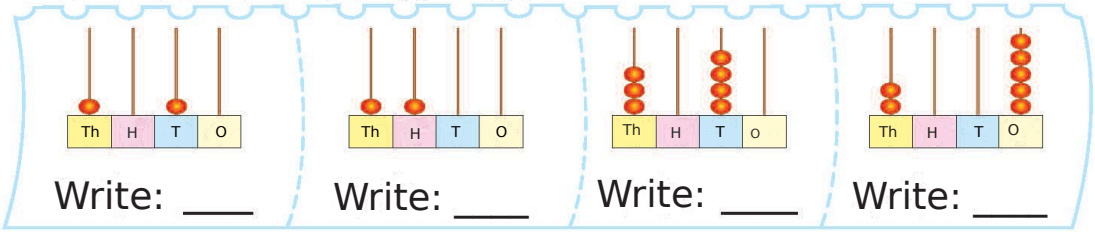
$5\square 7 < 523$　　$706 > \square 01$　　$32\square > 325$

3. 2436=2000+400+30+6. Look and fill in.



4. Count by hundreds from 8800 to 10000 and show each number on the counting frame.

5. Look, write and read.



6. Find the pattern and fill in.

(1) 388，389，_____，_____，392，393。

(2) 3260，3270，_____，3290，_____，3310。

(3) 5725，5825，_____，_____，6125。

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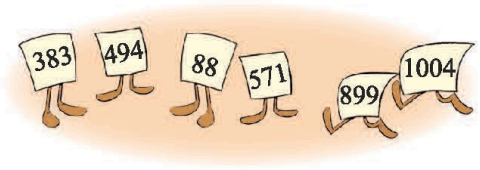
7. Think and talk.



(1) Which numbers are between 300 and 500?

(2) Which numbers are greater than 500?

(3) Which number is closest to 1000?

8.

Mark the approximate position of 2691.

The main peak of Changbai Mountain is about 2691 m above sea level.

9.

About ( ) beans.

About ( ) beans.

About 400 beans.

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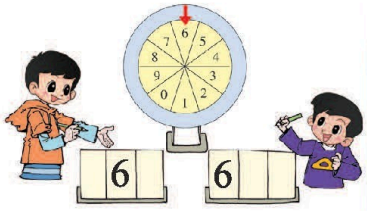
10. Maths game.



Game rules:

- Play in pairs. Each fills in their own boxes.
- Spin the spinner. When it stops, each player writes that digit in □□□ (any one box).
- Spin three times, and each makes a three-digit number.
- Compare them: the greater number wins.

(1) How many games did you and your partner play? How many did you win?

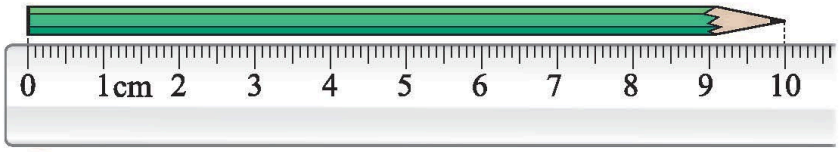
(2) In each game, what is the largest number you can get? And the smallest?

(3) Talk with your partner about how to win.

Unit 4 Measurement

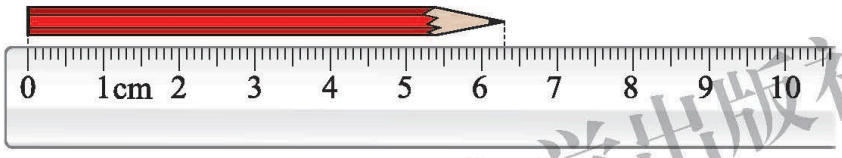
How Long Is the Pencil?

Measure how long each pencil is. Get to know the units.



It's exactly 10 cm.

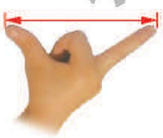
10 cm is 1 decimetre (dm).



6 cm plus 3 small spaces.

1 cm has 10 small spaces. Each small space is 1 millimetre (mm) long.

Find and talk.



1 hand span is about 1 dm long.

A 2-fen coin is about 1 mm thick.

Find 1 mm, 1 cm and 1 dm on the ruler. Fill in and think.

1 dm = ( ) cm 1 cm = ( ) mm 1 m = ( ) dm

Fill in and talk about it.

1 dm is 10 cm, so 3 dm is...

3 dm = ( ) cm
40 mm = ( ) cm

10 mm is 1 cm, so 40 mm is...

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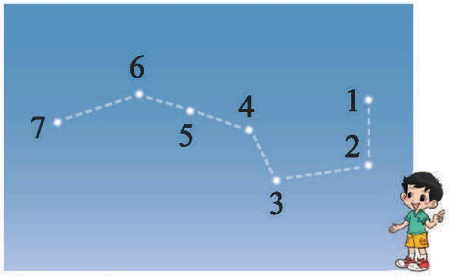
Practice

1. Estimate and measure.

	Estimate	Measurement
Pencil case length	_____	_____
Index finger length	_____	_____

2. This is a diagram of the Big Dipper.



Measure on the diagram: from star 1 to star 2 is ( ) mm, from star 6 to star 7 is ( ) cm

( ) mm

3. Fill in.

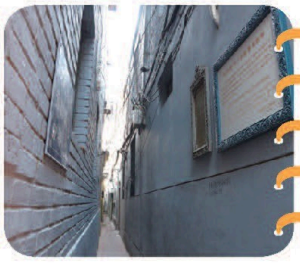
7 m = ☐ dm 3 cm = ☐ mm 90 dm = ☐ m

60 cm = ☐ dm 5 dm = ☐ cm 70 mm = ☐ cm

4. Read Taoqi's diary and correct the parts that don't make sense.

This morning I got up from my bed, which is 2 dm long. I brushed my teeth with a toothbrush 1 mm long, washed my face and dried it with a towel 70 m long. After breakfast I went to school. School is not far from my home, only about 900 mm away. At school, I sat on a stool 5 m high, took out my Chinese book, which is 7 cm thick, and got ready for the Chinese lesson.

5. Read and do.



The narrowest hutong in Beijing is Qianshi Hutong. It is 55 m long in total and 7 dm wide on average. With a partner, pull two desks apart so that they are 7 dm apart, then walk between them together.

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How Long Is 1 km?

How long is 10 m? Try it as shown.

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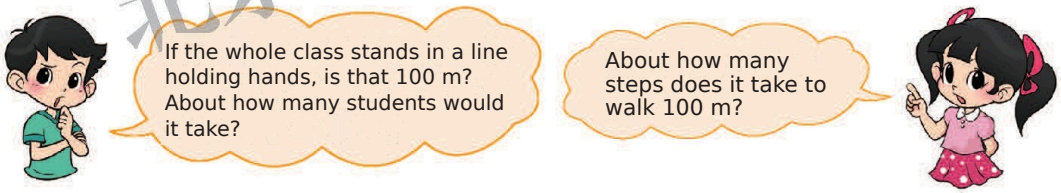
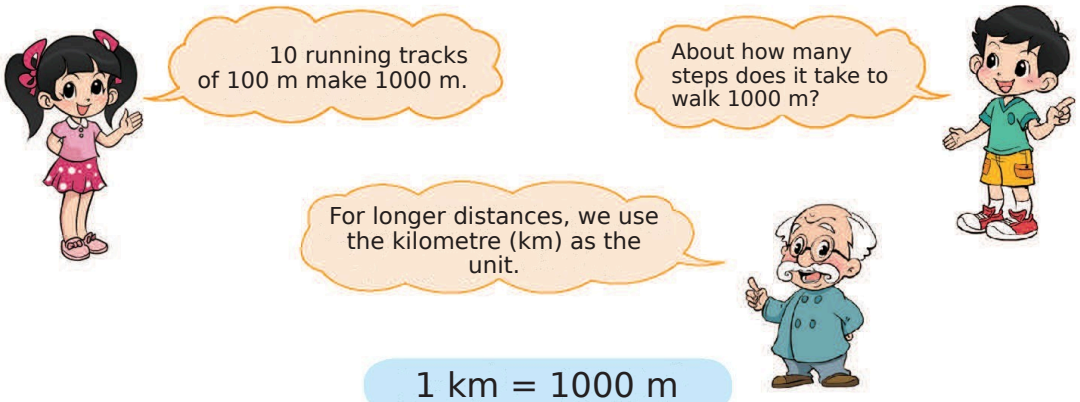
How long is 100 m? Think and do.



How long is 1000 m? Think, then discuss with a partner.



1 km = 1000 m

Practice

1. Write a suitable unit in each ( ).

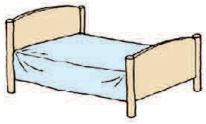
The tree is about 10 ( ).

In 1 hour the train goes about 120 ( ).

A marathon is about 42 ( ).

In 1 hour the plane flies about 800 ( ).

The bridge is about 400 ( ).



The bed is about 2 ( ).

2. Fill in.

4 km = ( ) m　2000 m = ( ) km　9 dm = ( ) cm

500 cm = ( ) m　30 mm = ( ) cm　7 cm = ( ) mm

3. Mini survey! What place is about 1 km from your home?

4. Clever Dog delivers letters.

(1) How many km does Clever Dog travel per hour?

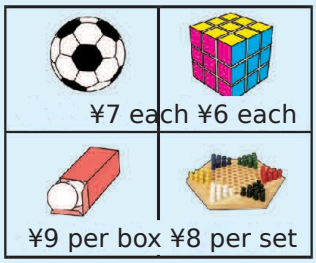
(2) At what time will Clever Dog probably reach the Forest Elder's home?

(3) If Clever Dog heads back the same way at 2:00 p.m., when might it get home?

Review

What I Learned

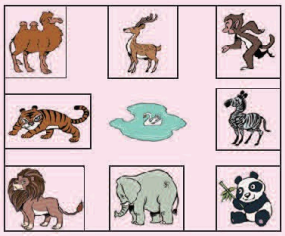
Buying Toys



How many footballs can ¥50 buy at most? What about other toys?

Share and calculate. Using how you shared, explain what each step of the column method means.

Animal Park



What is to the northeast of the pond? What about the other directions?

Think: which directions have you learned? Using the picture above, explain how you identify these directions.

Number of pupils

Guangming	1025
Yucai	1200
Qianjin	2406
Shengli	982
Huangcun	124

Which school has the most pupils?

Which place values have you learned? Use the counting frame and say what each "2" in the table above means.

On Duty

After school, we all cleaned the classroom together. I wiped the desks with a cloth 3 ( ) long, and Taoqi swept the floor with a broom 60 ( ) long. Soon the classroom was spotless. Lele pressed the switch 130 ( ) above the floor, locked the door 2 ( ) high, and everyone went home happily.

Review the length units you have learned and explain how they are related.

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My Progress

The "rulers" hidden on my body are...

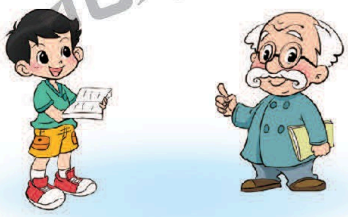
My favourite maths activity was using the direction board to record...

The big number that impressed me most in my survey is...

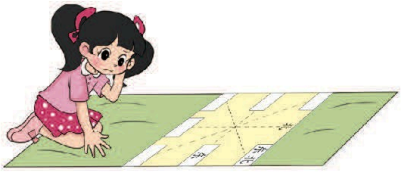
I can now solve division problems by drawing pictures.

My Questions

Why is column division different from column addition and subtraction?



Can every place between north and east be called the northeast direction?



How big is 1 hundred million? Is hundred millions the place after ten thousands?

The 6th national census shows that China's population is about 13.7 hundred million.

Is there a length unit smaller than the millimetre? Larger than the kilometre?

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Consolidate and Apply

1. 35 people entered the 100 m race on sports day, at most 8 people per group.

At least ☐ groups are needed. Explain each step of the column method.

2.

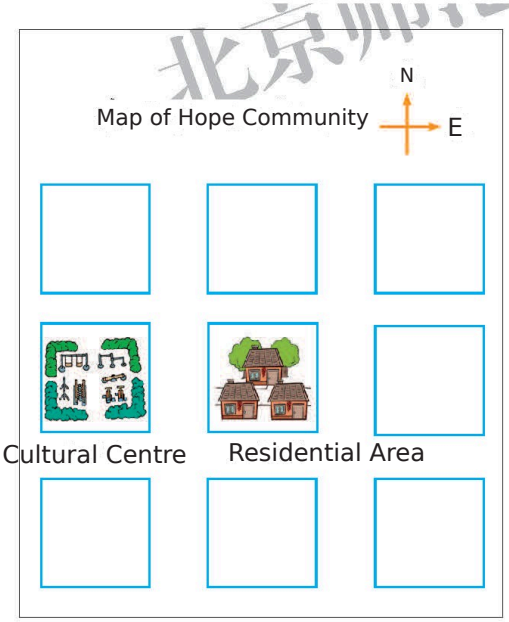
$3\overline{)28}$ $6\overline{)40}$ $7\overline{)52}$ $9\overline{)80}$

3. I am 15th in line. In which round will I get in?

4. One breakfast costs ¥4. How many breakfasts can you buy with ¥30 at most?

5. Cut and stick.

(1) Cut out Figure 2 from Appendix 1 and stick the pictures as described.



① The hill is southwest of the residential area.

② The shop is southeast of the residential area.

③ The clinic is east of the residential area.

④ Hope Primary School is west of the shop.

⑤ The experimental field is northwest of the residential area.

(2) Stick the pictures of the "fish pond" and the "chicken farm" on the map of Hope Community, and say which direction each one is in.

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6. Read, then write the underlined numbers in digits.

We can see about <u>three thousand</u> stars with the naked eye. ( )

The Pacific, the world's largest ocean, has an average depth of <u>four thousand and twenty-eight</u> metres. ( )

7. Count and fill in.

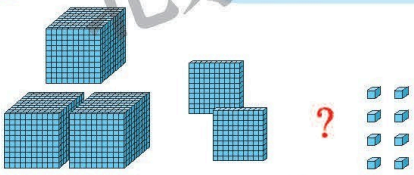
1 truck = 1000　1 basket = 100　1 bunch = 10

There are ( ) carrots in total.

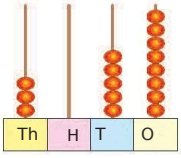
( ) sweets in total.

8. Think and fill in.

$3258 = 3000 + 200 + 50 + 8$



Need ( ) more



Hundreds rod: move ( )

9. Read, fill in and compare.

Chicken Farm Egg Record

Date	21	22	23	24
Eggs	908	1002	890	980

(1) ___ had the most eggs: ____.

(2) ___ had the fewest eggs: ____.

(3) Put the numbers of eggs laid on these days in order from greatest to least.

( ) > ( ) > ( ) > ( )

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10. About how many butterflies are there? Choose, then explain how you estimated.

10 butterflies

about 50	about 150	about 1500

11. Fill in.

4 dm = ___ cm　　3000 m = _____ km

6 km = _____ m　　8 cm = _____ mm

50 mm = _____ cm　　70 cm = _____ dm

12. Estimate, measure and talk.

	My estimate	My measurement
Length of my pencil	___ cm	___ cm ___ mm
Thickness of my book	___ mm	___ mm
Height of my stool	___ dm	___ dm ___ cm
Length of my favourite object		

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13.

How far is Lele's home from school?

Draw "○"

435 m	800 m	400 m

How far is Bumahu's home from school?

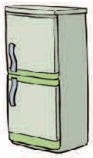
Draw "✓"

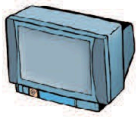
312 m	195 m	1 km

Unit 5 Addition and Subtraction

Buying Appliances

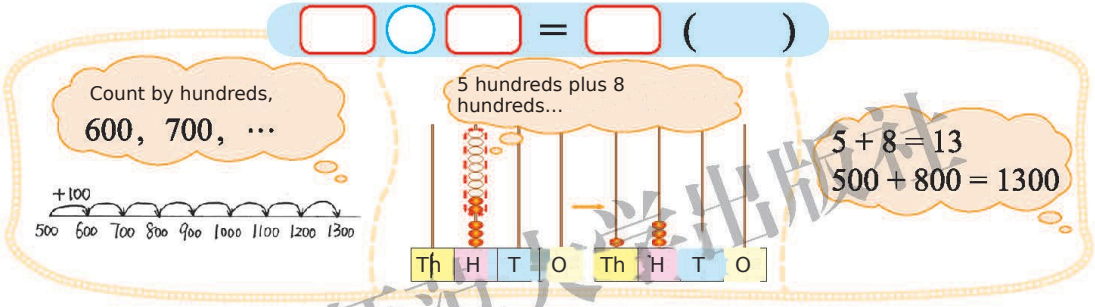
¥500


¥960


¥800

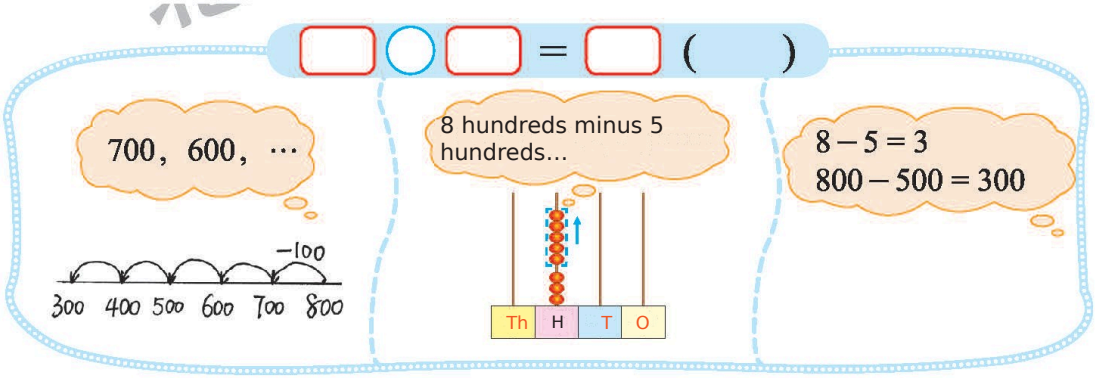
¥200

- How much do 1 washing machine and 1 television cost in total?



Answer: _____

- How much less does 1 washing machine cost than 1 television?



Answer: _____

- Ask another maths question and try to answer it.

How much do 1 refrigerator and 1 electric fan cost in total?

How much cheaper is 1 electric fan than 1 refrigerator?

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Practice

1. Draw, fill in, and move the beads on the counting frame.

$600 + 300 = \square$	$360 + 400 = \square$	$900 - 500 = \square$	$580 - 200 = \square$

2. $340 + 90 =$ $670 - 600 =$ $730 + 500 =$ $1800 + 200 =$

$340 - 90 =$ $670 - 60 =$ $730 - 500 =$ $1800 - 200 =$

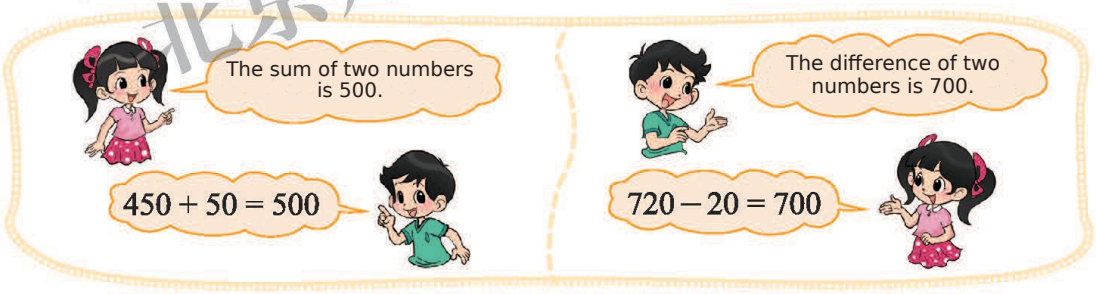
3. Lele wants to run a little farther each day. The table shows her runs for five days.

Mon	Tue	Wed	Thu	Fri
800 m	820 m	850 m	950 m	1150 m

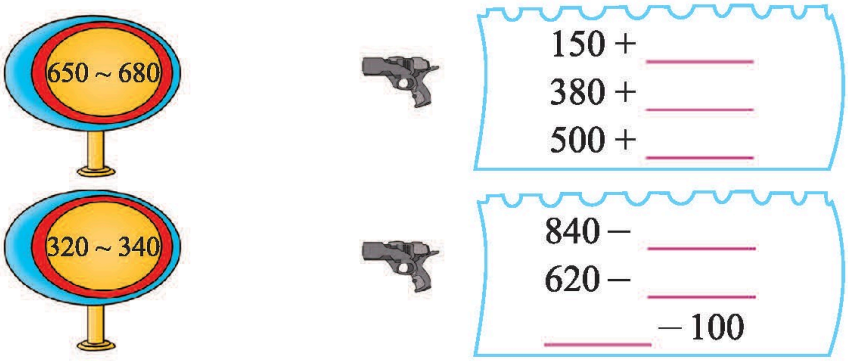
(1) How many more metres on Tuesday than Monday? Wednesday than Tuesday?

(2) Ask another maths question and try to answer it.

4. I say, you calculate



5. What numbers hit the target? Fill in two possibilities.



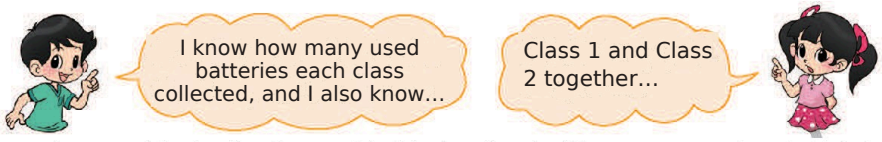
Recycling Batteries

Did You Know?

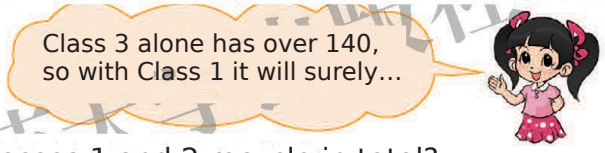
A used battery the size of a button can pollute as much water as a person drinks in a lifetime.

Prizes	Grade 2 battery collection	
200 batteries:	Class	Batteries
10 torches	Class 1	122
240 batteries:	Class 2	77
1 football	Class 3	149

Talk about the maths information in the picture and try to ask a maths question.



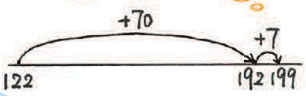
Estimate: which two classes together recycled enough batteries to win a prize?

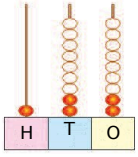


How many batteries did Classes 1 and 2 recycle in total?

( ) = ( )

122 + 70 = 192
192 + 7 = 199





Line up the same place values.

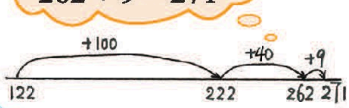
$$\begin{array}{r} 122 \\ +\ 77 \\ \hline 199 \end{array}$$

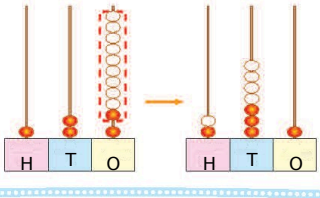
Answer: _____

How many batteries did Classes 1 and 3 recycle in total?

( ) = ( )

122 + 100 = 222
222 + 40 = 262
262 + 9 = 271





Make 10, carry 1.

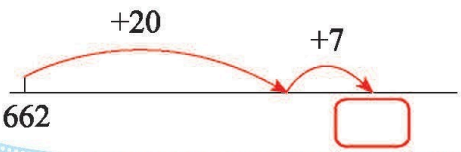
$$\begin{array}{r} 122 \\ +\ 149 \\ \hline 271 \end{array}$$

Answer: _____

Practice

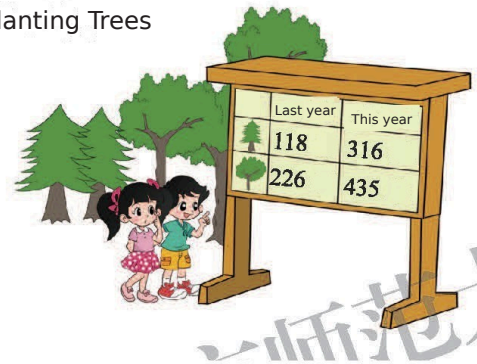
1. Draw, fill in, and move the beads on the counting frame.

$662 + 27 = \square$



$453 + 129 = \square$

2. Planting Trees



(1) What maths information can you see in the picture? Talk about it with your partner.

(2) Estimate: did the forest farm plant at least 800 trees this year?

(3) How many trees in all this year?

(4) In the 2 years, how many trees in all? And trees?

3. Use the column method.

$245 + 172 =$ $433 + 158 =$

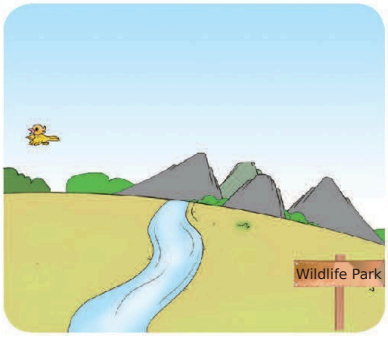
$166 + 329 =$ $306 + 275 =$

4. The school is 360 m east of the post office, and Xiaoxiao's home is 264 m west of it.

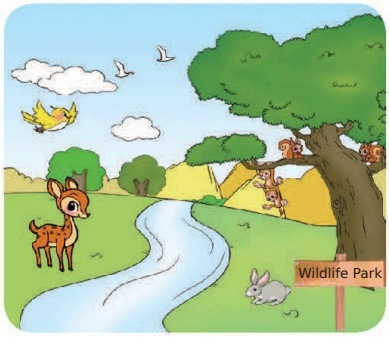
(1) Mark the school and Xiaoxiao's home roughly on the diagram.

(2) Work out how many metres it is from Xiaoxiao's home to the school.

Ten Years of Change



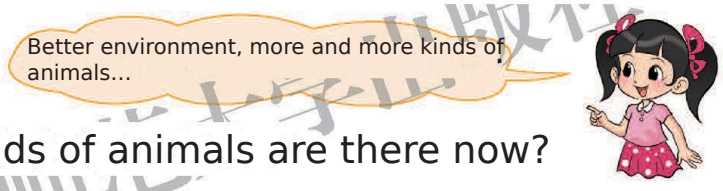
Ten years ago



Now

Ten years ago, Qingshan County Wildlife Park had only 87 kinds of animals. Since the animals' habitat was improved, 139 more kinds of animals have come to live here.

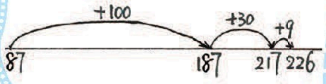
Talk about how the kinds of animals in the Wildlife Park have changed over ten years.

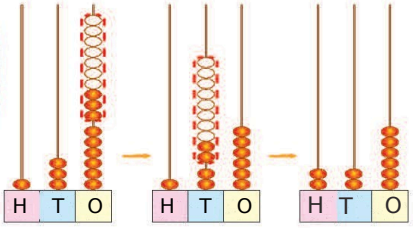


How many kinds of animals are there now?

□ ○ □ = □ ( )

87 + 100 = 187
187 + 30 = 217
217 + 9 = 226





If a place adds up to ten or more, carry 1 to the next place.

$$\begin{array}{r} 8\ 7 \\ +\ 1_1 3_1 9 \\ \hline 2\ 2\ 6 \end{array}$$

Answer: _____

Use the column method.

Think: what must you be careful about when adding with the column method?

268 + 89 =

304 + 197 =

686 + 235 =

Practice

1. Draw, fill in, and move the beads on the counting frame.

389 + 93 = ☐

389

453 + 179 = ☐

453

2.

¥368

¥68

¥365

¥580

(1) How much is 1 desk and chair set?

(2) How much do 1 bed and 1 bookcase cost in total?

(3) Which two can you buy with ¥800?

3. Use the column method.

487 + 58 =

337 + 584 =

199 + 726 =

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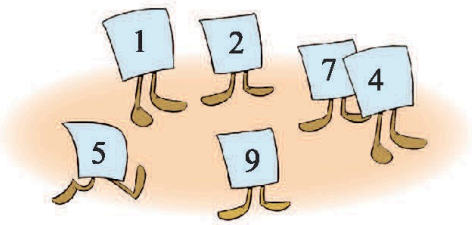
4. Forest Doctor

$$\begin{array}{r} 397 \\ +\,432 \\ \hline 729 \end{array}$$

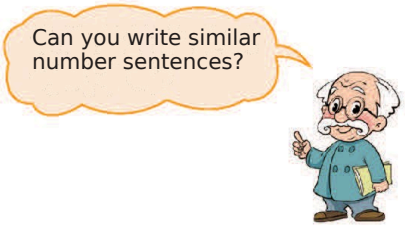
$$\begin{array}{r} 435 \\ +\,275 \\ \hline 700 \end{array}$$

$$\begin{array}{r} 278 \\ +\,256 \\ \hline 22 \end{array}$$

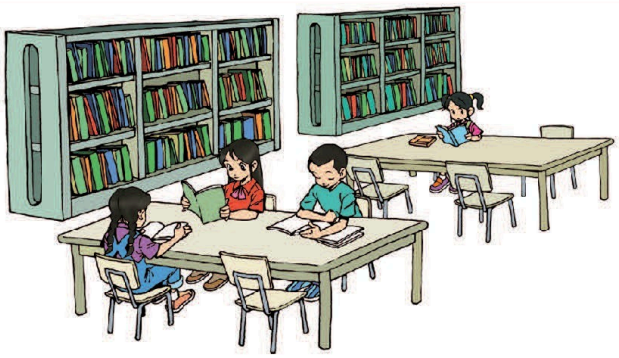
5. Use 6 cards to form two three-digit numbers so that their sum is 1000.



$$\begin{array}{r} 471 \\ +\,529 \\ \hline 1000 \end{array}$$



Little Library



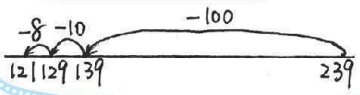
Books in the Little Library

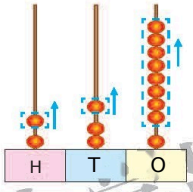
Type	Books
Stories	239
Comics	118
Science	57

How many more storybooks than comics are there?

( )

239 − 100 = 139
139 − 10 = 129
129 − 8 = 121





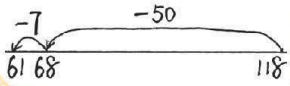
$$\begin{array}{r} 239 \\ -\ 118 \\ \hline 121 \end{array}$$

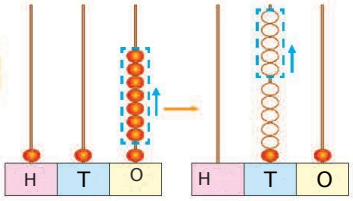
Answer: _____

How many fewer science books than comics are there?

( )

118 − 50 = 68
68 − 7 = 61

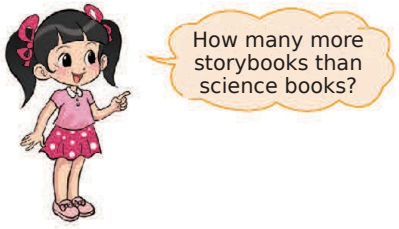




$$\begin{array}{r} \dot{1}\,18 \\ -\ 57 \\ \hline 61 \end{array}$$

Answer: _____

Make up a problem that 239 − 57 can solve, and work it out.



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Practice

1. Draw, fill in, and move the beads on the counting frame.

$266 - 124 = \square$

$592 - 254 = \square$

2. Use the column method.

$478 - 236 =$ $763 - 128 =$

$852 - 436 =$ $385 - 179 =$

3. Think and calculate.

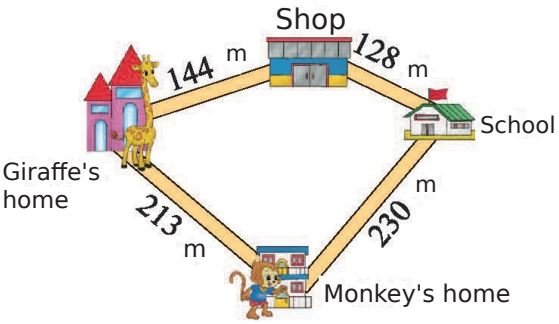
$300 + \square = 580$ $\square + 80 = 320$

$560 - \square = 350$ $\square - 470 = 240$

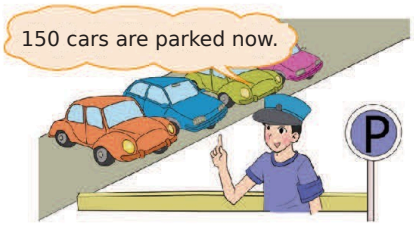
4. Animals go out.

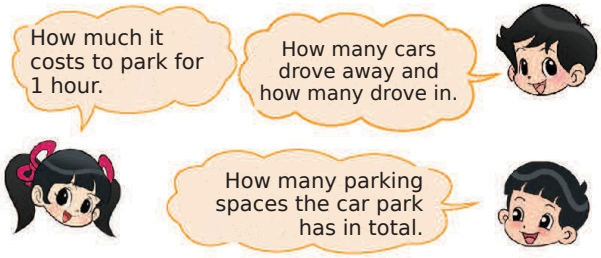
(1) Which way is shorter for the giraffe to go from home to school? How many metres must it walk?

(2) Which way is shorter for the monkey to go from home to the shop? How many metres shorter?



5. To find out how many cars will be in the car park in 1 hour, what else do you need to know? Think about it: who is right?





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Exercise 3

1. Draw, fill in, and move the beads on the counting frame.

$149 + 223 =$ ☐

149

$312 - 130 =$ ☐

312

2. Write >, < or = in each ○.

$280 + 300 \bigcirc 280 + 30$ $\quad 6400 - 400 \bigcirc 6400 - 400$

$750 + 40 \bigcirc 40 + 750$ $\quad 900 - 80 \bigcirc 960 - 80$

3. Use the column method.

$498 + 39 =$ $\quad 105 + 789 =$

$366 - 214 =$ $\quad 765 - 127 =$

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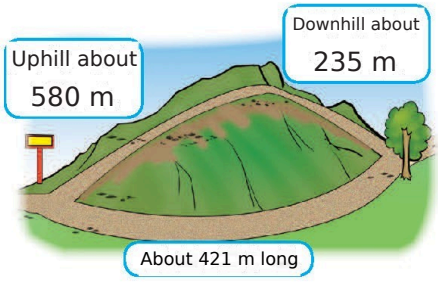
4. Stamp collecting.

(1) How many stamps did the two groups collect altogether?

(2) How many fewer stamps did Xiaoxiao's group collect than Taoqi's group?

5.



(1) How many metres up and down altogether?

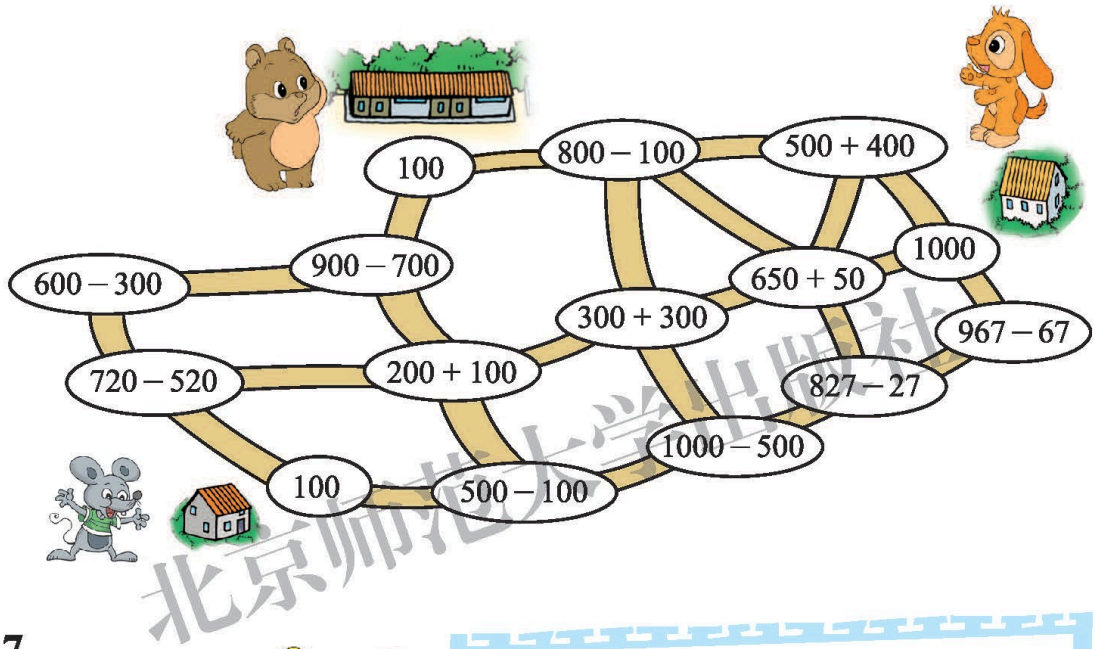
(2) How many metres longer is the mountain path than the flat road?

(3) How many metres shorter is the way down than the way up?

6. Go through the maze. Trace each little animal's route as instructed.

(1) The bear must follow the path with the answers 100, 200, 300, ..., 1000 to reach the dog's home.

(2) The dog must follow the path with the answers 1000, 900, 800, ..., 100 to reach the mouse's home.



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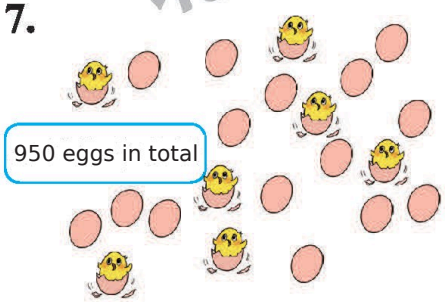
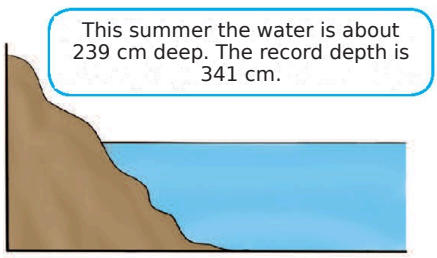
7.



(1) 227 chicks hatched in the morning. How many eggs have not hatched yet?

(2) In the afternoon 218 more chicks hatched than in the morning. How many chicks hatched in the afternoon?

8.

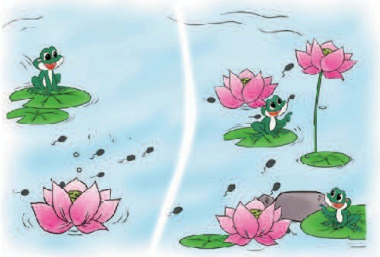


(1) How many centimetres short of the record depth is the water this summer?

(2) This winter, the river's water depth is about 110 cm less than in summer. About how many centimetres deep is the water in winter?

How Tadpoles Grow

Did you know? Out of 100 tadpoles, usually only about 20 become frogs.

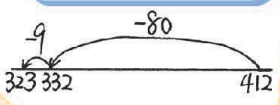


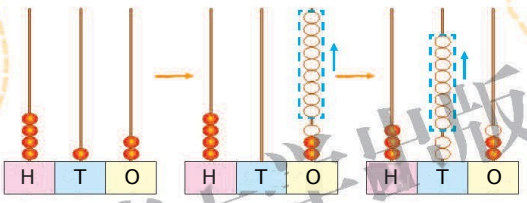
	West Pond	East Pond
Tadpoles at first	514	412
Became frogs	126	89

- How many tadpoles in East Pond did not become frogs?

□ ○ □ = □ ( )

412 − 80 = 332
332 − 9 = 323





$$\begin{array}{r} 4\,1\,2 \\ -\;\;8\,9 \\ \hline 3\,2\,3 \end{array}$$

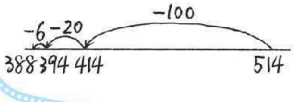
Why isn't it 4?

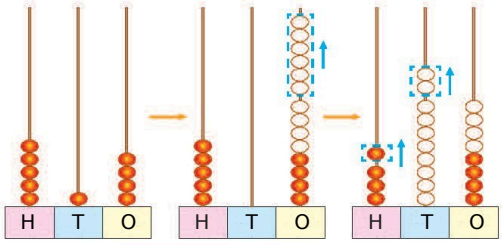
Answer: _____

- How many tadpoles in West Pond did not become frogs?

□ ○ □ = □ ( )

514 − 100 = 414
414 − 20 = 394
394 − 6 = 388





$$\begin{array}{r} 5\,1\,4 \\ -\;1\,2\,6 \\ \hline 3\,8\,8 \end{array}$$

Answer: _____

- Are they right? Talk with a partner.

$$\begin{array}{r} 5\,2\,6 \\ -\;4\,1 \\ \hline 1\,1\,6 \end{array}$$

$$\begin{array}{r} 4\,8\,2 \\ -\;1\,7\,8 \\ \hline 3\,1\,4 \end{array}$$

Be careful when you calculate!

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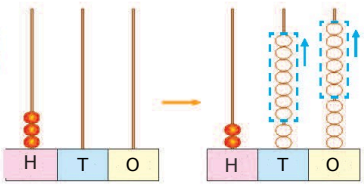
Try It

We bought 300 balloons for the party and have blown up 76. How many are still not blown up?

□ ○ □ = □ ( )

$300-70=230$
$230-6=224$



The tens digit is 0. What now?

$$\begin{array}{r} 300 \\ -\ 76 \\ \hline 224 \end{array}$$

Think: what should we watch out for in column subtraction?

Answer: _____。

Xiaoxiao calculated like this. Do you understand it? Tell a partner.

$300=299+1$
$299-76=223$
$223+1=224$

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Practice

1. Draw, fill in, and move the beads on the counting frame.

Talk with a partner: is each step correct?

$513-46=$□

513

$632-278=$□

632

2. How many pages are left to read?

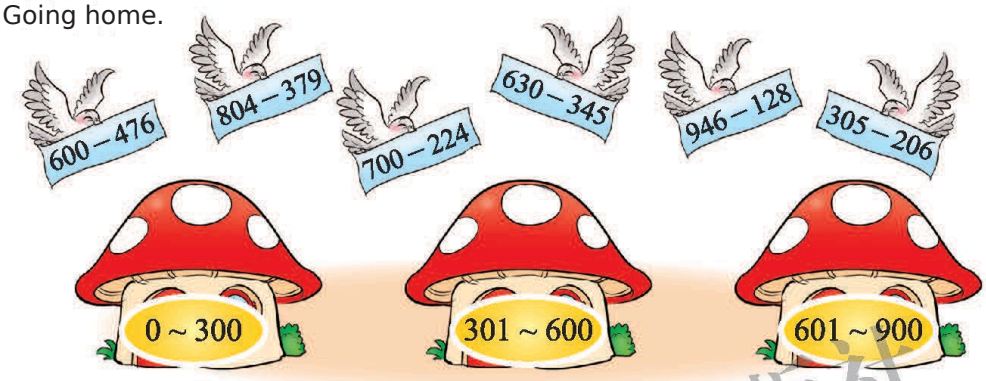
215 pages in all

I've read 78 pages

3. Use the column method.

$125-36=$	$210-145=$	$331-138=$	$900-229=$

4. Going home.

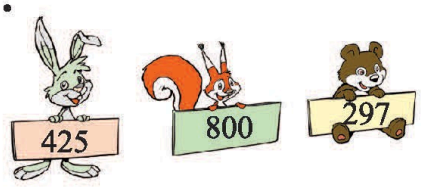


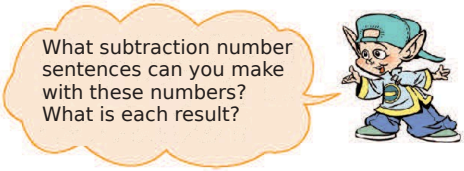
5. Fill in.

$102 = 99 + \square$ $205 = 199 + \square$ $508 = 499 + \square$

6. How much change?

7.





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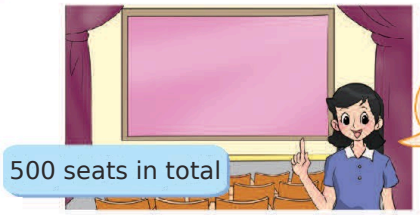

The school is taking Grade 1 and Grade 2 students to see a film.

Grade 1	203 students
Grade 2	278 students

(1) Estimate: are there enough seats in the theatre?

(2) Calculate: how many students are going to see the film in all?

(3) Are there enough seats? What is the difference between the number of seats and students?

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Is It Correct?

How much should they pay? Did the saleswoman calculate correctly?

□ ○ □ = □ ( )

I worked it out again. It's ¥256.

$$\begin{array}{r} 138 \\ +\ 118 \\ \hline 256 \end{array}$$

I did 256 minus 118 and got exactly 138. It's right.

$$\begin{array}{r} 256 \\ -\ 118 \\ \hline 138 \end{array}$$

This is called checking. It can help us make fewer mistakes.

Answer: _____。

They paid the saleswoman ¥300 and got ¥44 change. Is that right?

I'll try 300 minus 44.

$$\begin{array}{r} 300 \\ -\ 44 \\ \hline 256 \end{array}$$

I'll use addition to check...

$$\begin{array}{r} 256 \\ +\ 44 \\ \hline 300 \end{array}$$

Answer: _____。

Calculate and check.

$367 + 254 =$　　$732 - 186 =$

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Try It

Spotting calculation mistakes early and finding the reasons is a very good way to learn.

Addition errors		Subtraction errors	
$234+256=480$	$296+153=431$	$515-49=474$	$900-632=368$
$\begin{array}{r} 234 \\ +\ 256 \\ \hline 480 \end{array}$	$\begin{array}{r} 296 \\ +\ 135 \\ \hline 431 \end{array}$	$\begin{array}{r} 515 \\ -\ 49 \\ \hline 474 \end{array}$	$\begin{array}{r} 900 \\ -\ 632 \\ \hline 368 \end{array}$

What went wrong? Calculate and explain.

$\begin{array}{r} 234 \\ +\ 256 \\ \hline 480 \end{array}$ forgot to...

$\begin{array}{r} 900 \\ -\ 632 \\ \hline 368 \end{array}$ when borrowing twice in a row...

Discuss: what are some good ways to make fewer calculation mistakes?

Checking can reduce mistakes.

Remember to carry and borrow.

Woof! Be careful not to copy the numbers wrong!

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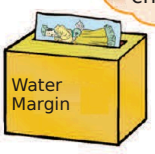
Practice

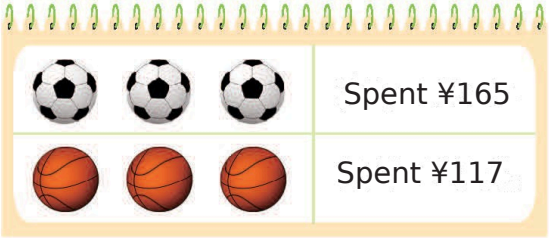
1. How many cards did Taoqi and Xiaoxiao collect in total? Calculate and check.

I collected 136 Three Kingdoms character cards.



I collected 288 Water Margin character cards.

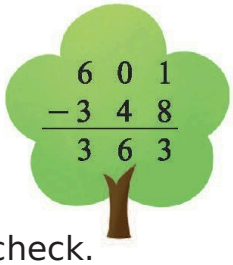
2.

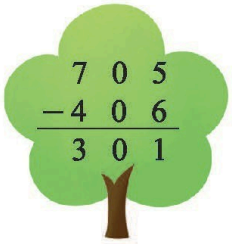


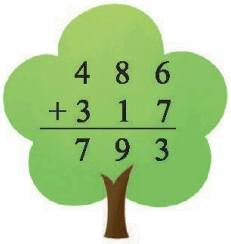
(1) How much was spent in total?

(2) Teacher Li paid ¥300 and got ¥28 change. Is that right?

3. Forest Doctor







4. Calculate and check.

$241 + 672 =$ $564 - 146 =$

$368 + 475 =$ $724 - 139 =$

5. Making shell crafts

(1) How many more shells does a sheep use than a deer?

(2) How many fewer shells does a sheep use than a rooster?

(3) Check the results and say how you checked them.

(4) Pose another maths question and try to answer it.

6. In PE class.

Long jump

Taoqi	235 cm
Qisi	189 cm
Maths Fan	216 cm

High jump

Xiaoliang	105 cm
Xiaogang	95 cm
Xiaoqiang	101 cm

(1) How much farther did Taoqi jump than Qisi?

(2) How much higher did Xiaoliang jump than Xiaogang?

(3) Pose another maths question and try to answer it.

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Exercise 4

1. Draw, fill in, and move the beads on the counting frame.

$862 + 56 = \square$ $534 + 229 = \square$ $625 - 131 = \square$

862 534 625

2. Sunshine Shoe Shop's sales. Ask questions about the chart and answer them.

3.

$50 - 16 =$ $380 - 250 =$ $1000 - 800 =$

$63 + 52 =$ $460 + 70 =$ $700 - 190 =$

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4. Calculate and check.

$78 + 280 =$ $718 - 542 =$ $418 - 159 =$

5. Put these number sentences in order of their answers, smallest first.

_____ < _____ < _____ < _____

6. Making one toy bicycle takes 186 cm of wire.

(1) Estimate: is there enough wire left to make another toy like this one?

(2) How many centimetres of wire are left?

7. Calculate: how much cheaper is each of these items than before?

E-dictionary	Cassette player	Camera
Was: ¥398 Now: ¥350	Was: ¥235 Now: ¥198	Was: ¥1000 Now: ¥960

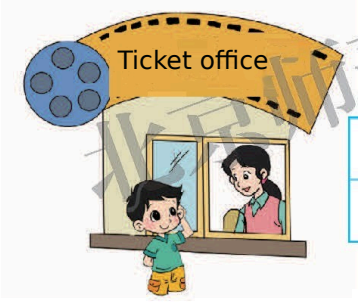
8.

From experience, each year we bring in 25 fewer boxes of apples in March than in February.

(1) How many boxes are needed in March?

(2) How many boxes in all for both months?

9.



Guangming Cinema's ticket sales for one week

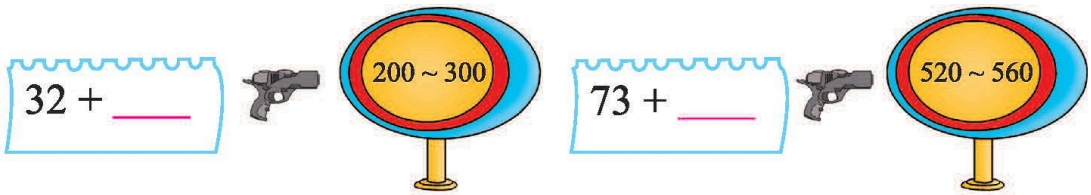
Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Tickets	156	395	162	214	480	402	465

(1) Which day had the most tickets sold? Which day the fewest? What is the difference?

(2) Which two days sold the most tickets in total? And the fewest?

(3) Which two days have the closest ticket sales?

10. What number hits the target? Give two answers.

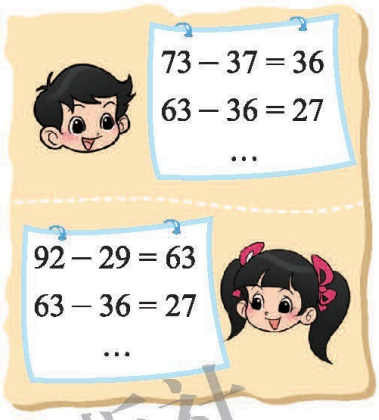


11. The Number Castle is the home of the Wise Old Man. It has two gates with combination locks. Can you open them? Calculate and discuss.

Gate 1

Code hint: Pick any two different digits from 0 to 9. Make the largest and the smallest numbers with them and subtract. Keep going like this, as shown below, and you will find the code for the first gate.

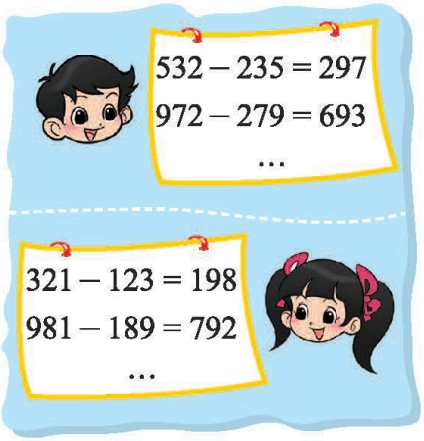
$$81 - 18 = 63$$
$$63 - 36 = 27$$
$$72 - 27 = 45$$
$$\cdots$$

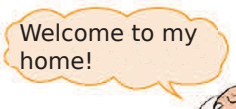


Gate 2

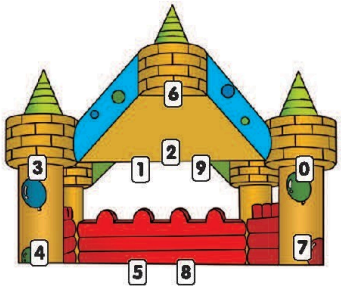
Code hint: Pick any three different digits from 0 to 9. Make the largest and the smallest numbers with them and subtract. Keep going like this, as shown below, and you will find the code for the second gate.

$$972 - 279 = 693$$
$$963 - 369 = 594$$
$$\cdots$$





In the world of maths there are many, many amazing numbers like these.



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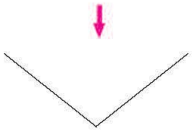
Unit 6 Recognising Shapes

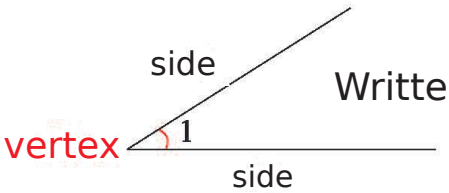
Recognising Angles

In everyday life, we often see all kinds of angles.



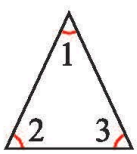
Draw.

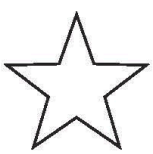
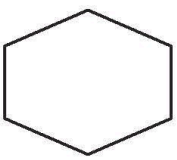
Let's learn.



Written as: ∠1. Read as: angle 1.

Find three angles in each figure below and mark them.



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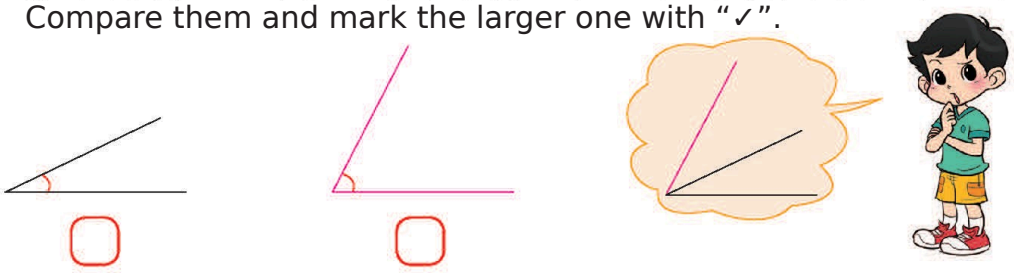
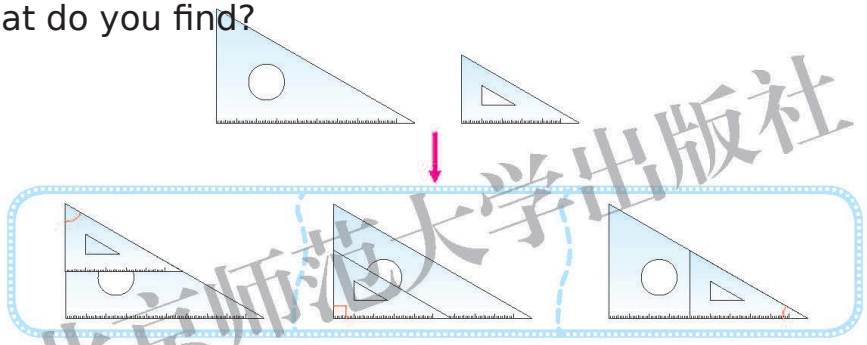
Try It

- Cut out the two angles in Figure 1 of Appendix 3. Compare them and mark the larger one with "✓".



- As shown, compare the sizes of the angles of the two set squares. What do you find?



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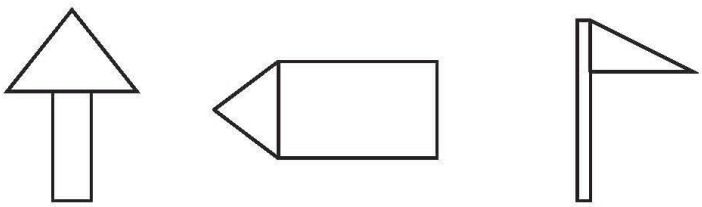
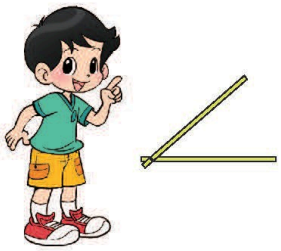
Practice

1. Use a set square to draw an angle, then trace its sides and vertex.

2. Mark three angles in each figure below.



3. This is the movable angle Taoqi made. Make one yourself and try it.



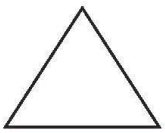
(1) How can you make Taoqi's angle smaller?

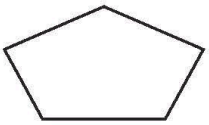
Do it and talk about it.

(2) How can you make Taoqi's angle larger?

Do it and talk about it.

4. Count. How many angles does each shape below have?





( ) angles ( ) angles ( ) angles

5. Calculate and check.

$236 + 624 =$ $805 - 167 =$

$192 + 275 =$ $500 - 381 =$

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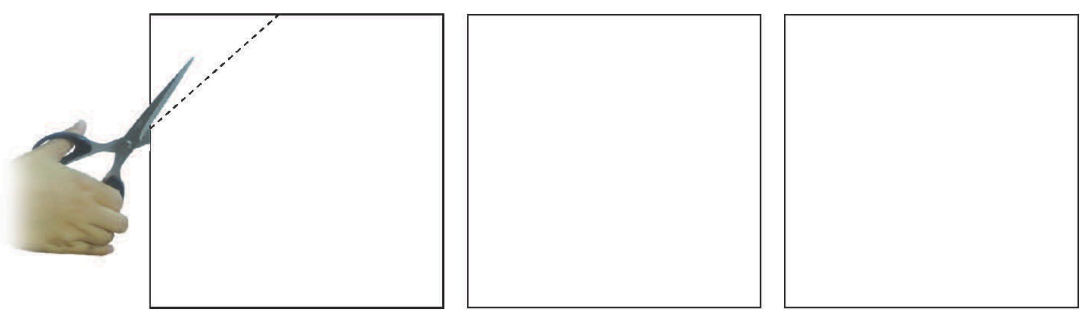
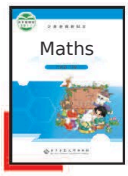
6. Do it and fill in. A square piece of paper has 4 angles. If you cut off one angle with scissors, how many angles are left?



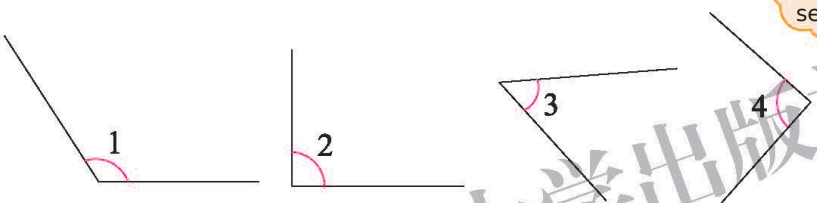
( ) angles left ( ) angles left ( ) angles left

Right Angles



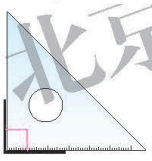
These are all right angles.

Say which of the angles below are right angles.

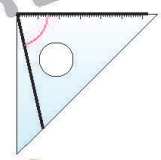


I'll compare them with a set square.

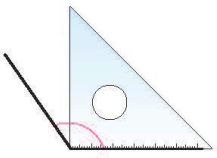
Compare and learn.



A right angle.

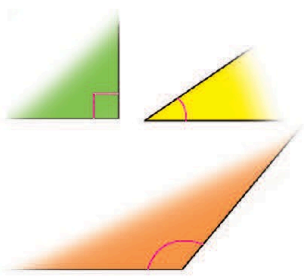


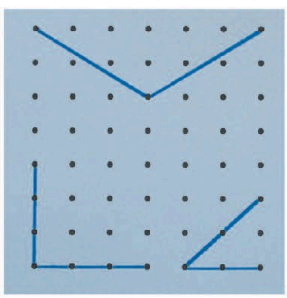
Smaller than a right angle: an acute angle.



Larger than a right angle: an obtuse angle.

Fold and draw.





Which objects around you have right, acute or obtuse angles?

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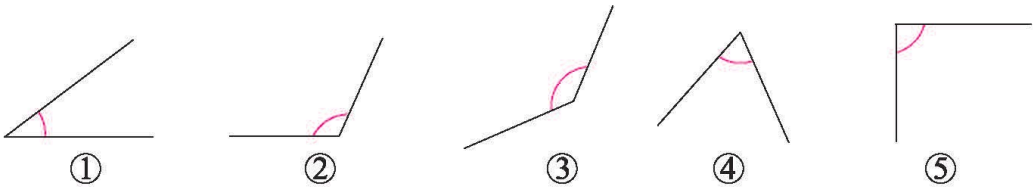
Practice

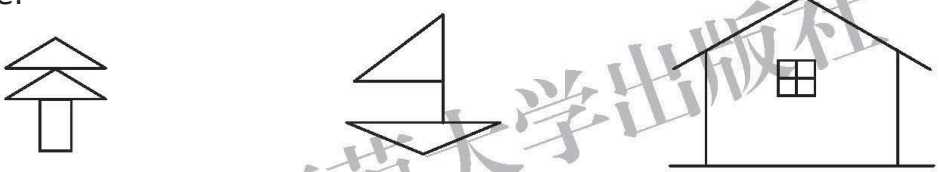
1. Use sticks to make 1 right angle, 1 acute angle and 1 obtuse angle.

2. Use a set square to compare and fill in.



Acute _____ Right _____ Obtuse _____

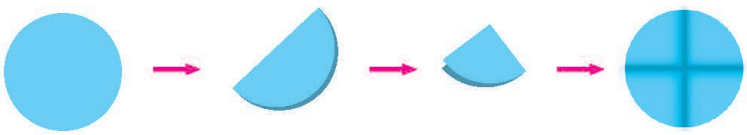
3. In each figure, mark 1 right angle, 1 acute angle and 1 obtuse angle.



4. Calculate and check.

$612-224=$ $245+176=$ $438+371=$

5. Fold a circle in half twice, then open it. Discuss where the right angles are.

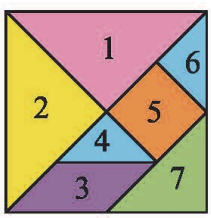


Fold Fold again Unfold

6. Look and do.

(1) Which tangram piece has a right angle? Which piece has an obtuse angle?

(2) Use two tangram pieces from Figure 2 of Appendix 3 to make a right angle and an obtuse angle.



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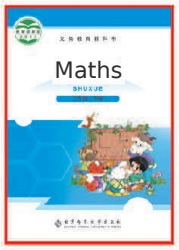
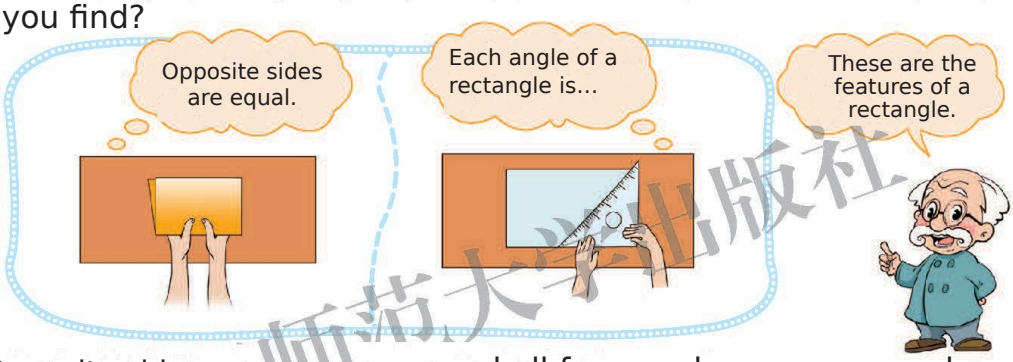
Rectangles and Squares



Rectangle

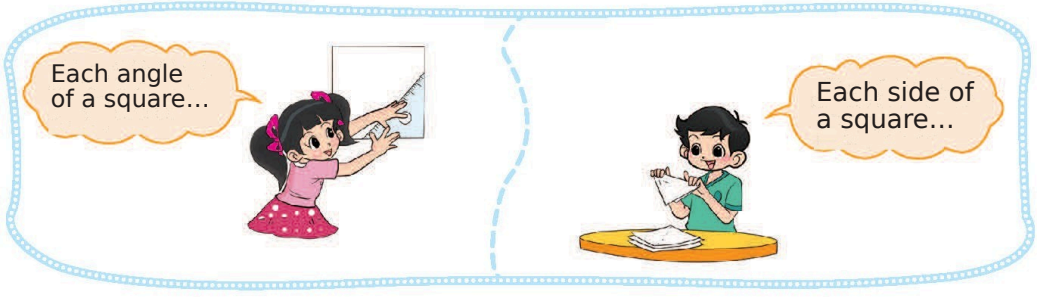
Square

What are the features of a rectangle? Fold it and measure it. What do you find?



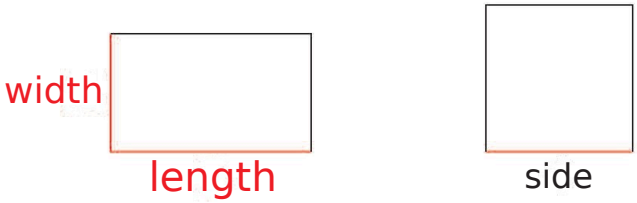
Opposite sides are_____ and all four angles are ____ angles.

What are the features of a square? Fold it, measure it and fill in.



The four sides are _____, and all four angles are ____ angles.

Let's learn.



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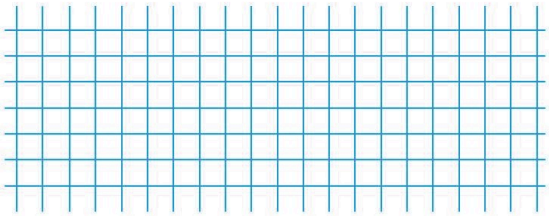
Practice

1. Draw a square and a rectangle on the grid paper, and discuss their features.

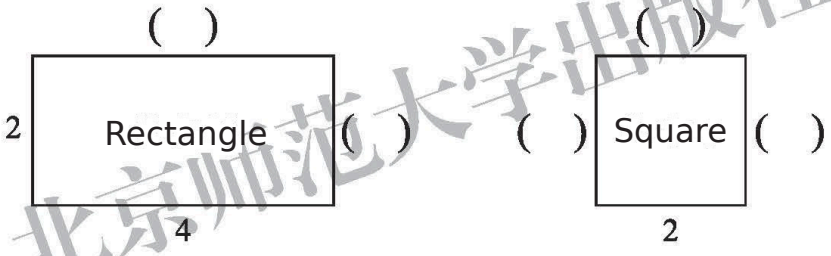


2. Do it.

(1) Which tangram pieces can make a rectangle or a square? Try it with the tangram in Figure 2 of Appendix 3.

(2) Use sticks to make a rectangle and a square.

3. Fill in the missing numbers. (Unit: cm)



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4. Use a rectangular piece of paper 12 cm long and 8 cm wide to fold a square. What is the side length of the largest square?

5. Calculate and check.

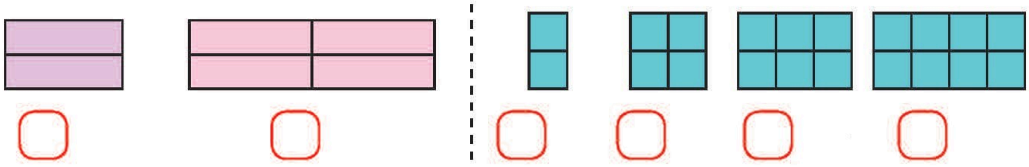
$199 + 398 =$ $263 - 26 =$ $600 - 365 =$

6. Count and fill in.

(1) How many rectangles?

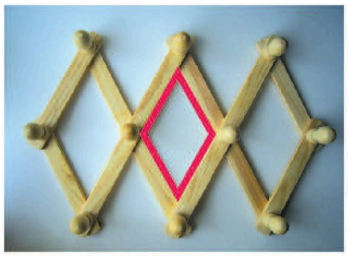
(2) How many squares?

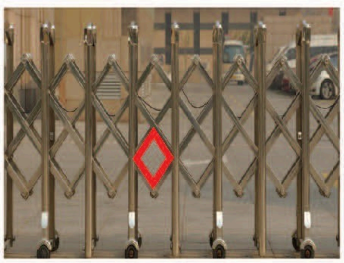
What do you notice?

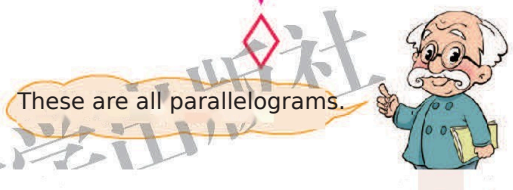


Parallelograms

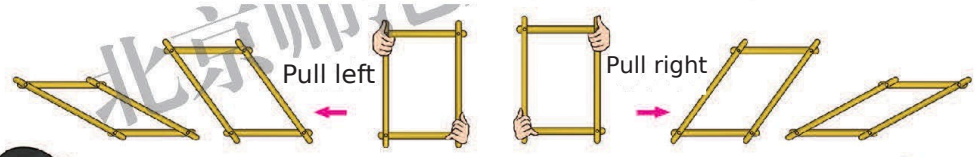
Look and recognise. In daily life, we often see parallelograms.

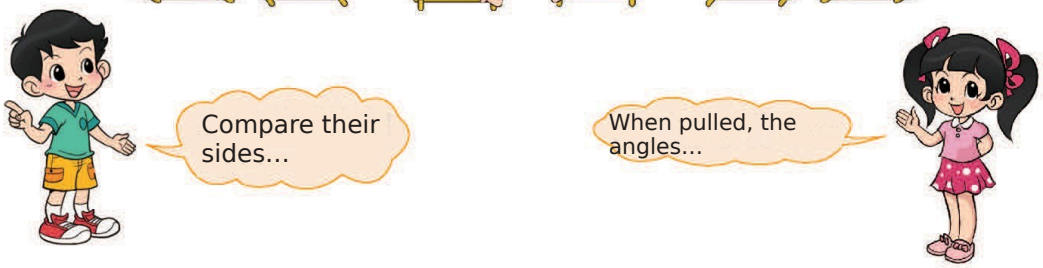




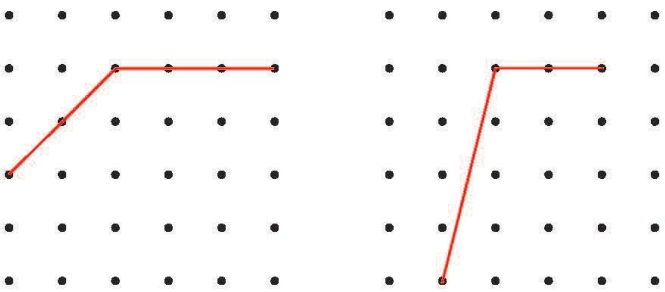


Do it and discuss: what has changed in this shape, and what has not?





Finish drawing the parallelograms.



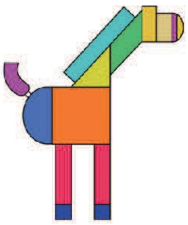
Practice

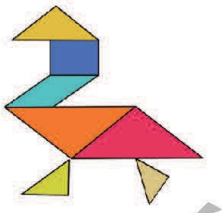
1. Do it.

(1) Make a parallelogram with the tangram (Appendix 3, Figure 2).

(2) Use sticks to make a parallelogram.

2. In each pattern below, find three shapes you know.

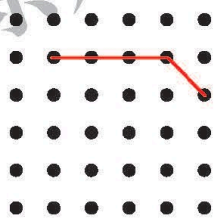


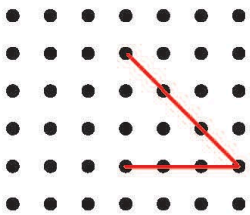


3. Calculate and check.

$327 + 143 =$ $627 - 343 =$ $800 - 126 =$

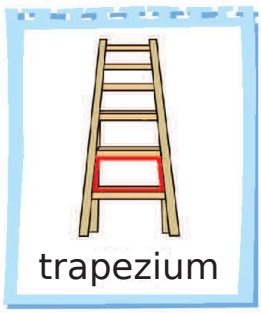
4. Finish drawing the parallelograms.

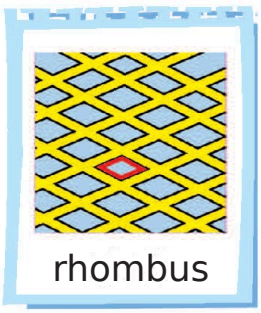


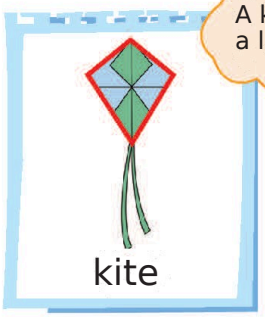


5. Read and tell.

Life is full of interesting quadrilaterals.







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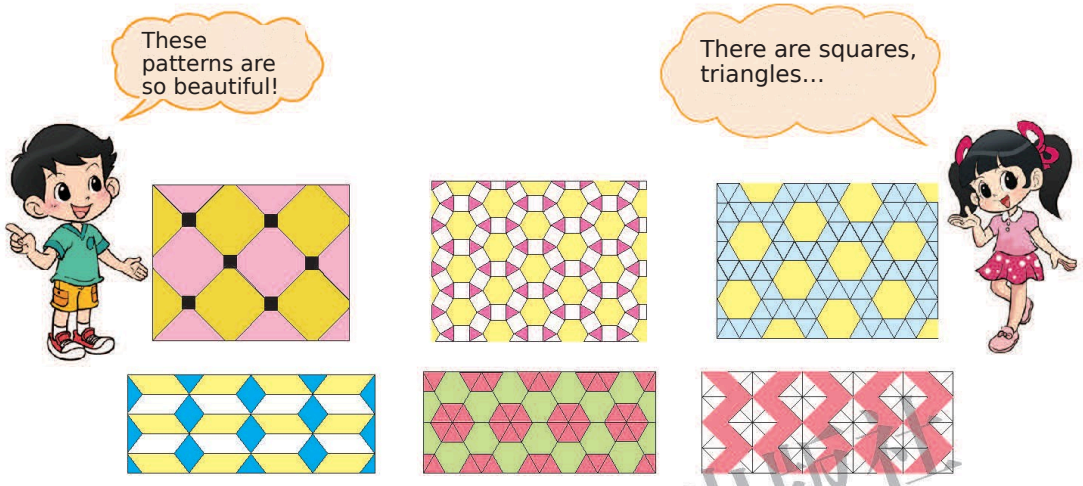
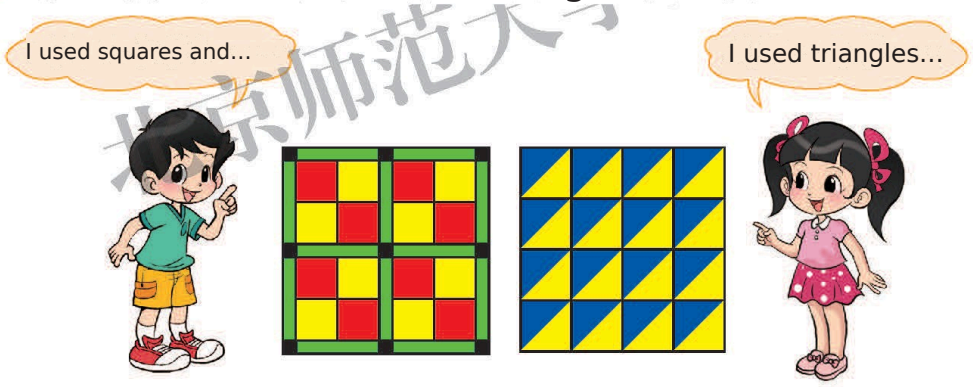
Enjoy and Design

- Look at the patterns below and find the shapes you know.



- Talk about how the two designs below were made.



- Try designing a beautiful pattern of your own on the grid paper.

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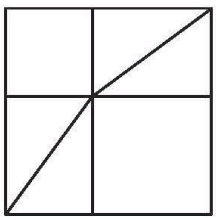
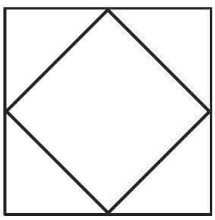
Practice

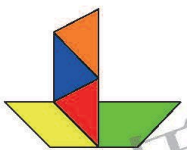
1. Find two shapes you know in each figure below, trace them and talk about them.





2. Look and do.

Find 3 parallelograms and trace them.



Use 4 rectangles to make a new shape.

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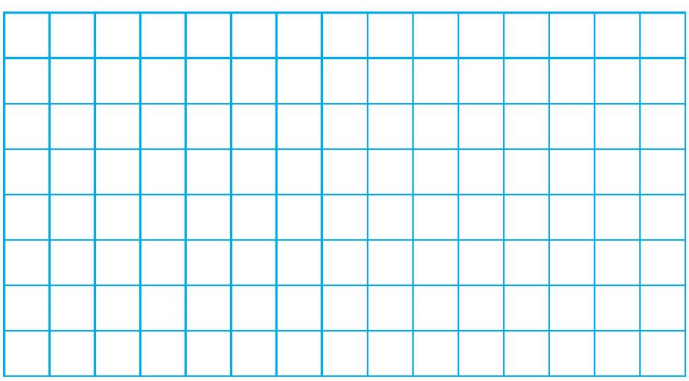
3. Calculate and check.

$125 + 743 =$ $362 + 415 =$

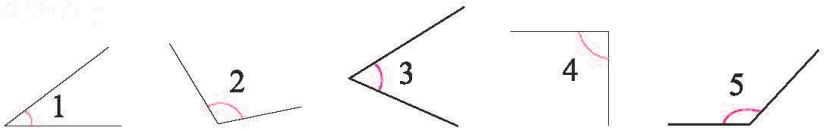
$700 - 449 =$ $502 - 238 =$

4. Pick your favourite of your classmates' patterns and trace it on the grid paper below.



Exercise 5

1. Use a set square to tell what kind of angle each one below is, and name the parts of one angle.

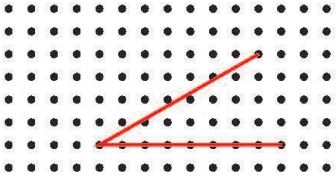


2. Say which angles in the shapes below are acute, which are right and which are obtuse.

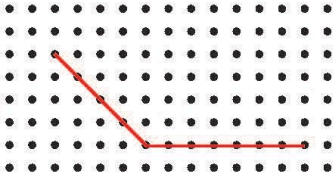
3. Can you name two times when the hour and minute hands make angles of the same size?

4. Think and draw.

Draw a smaller acute angle.

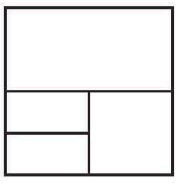


Draw a larger obtuse angle.

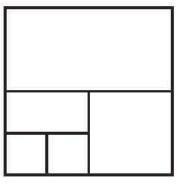


5. Find and trace.

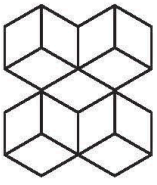
Trace 3 rectangles.



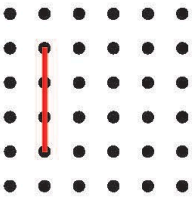
Trace 3 squares.



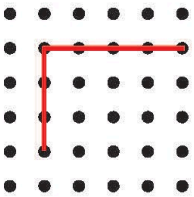
Trace 3 parallelograms.



6. (1) Finish drawing the square.

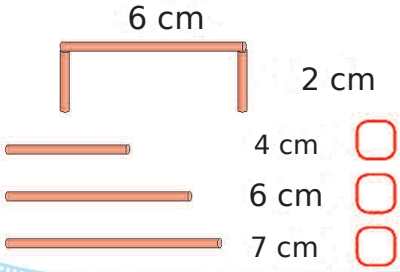


(2) Finish drawing the rectangle.

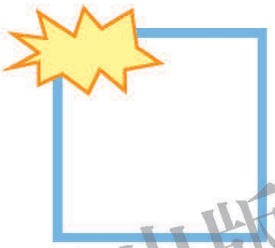


7. Think, then tick (✓) the correct answer.

Which stick can be used to form a rectangle?



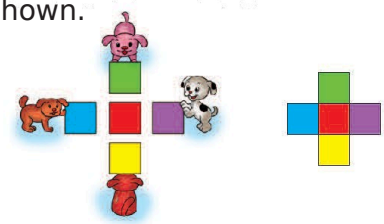
The hidden angle of the square is:



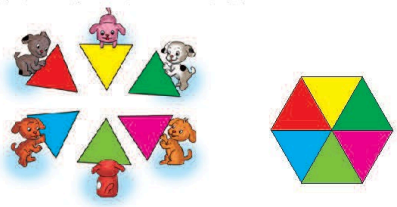
acute ☐
right ☐
obtuse ☐

8. Look and do. What else can you make?

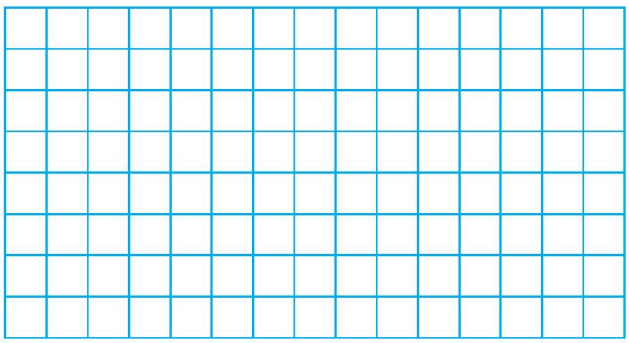
Make 5 squares from cardboard and put them together as shown.



Make 6 triangles from cardboard and put them together as shown.



9. Design a pattern you like on the grid paper.



Unit 7 Hours, Minutes and Seconds

Olympic Opening

At 8:08 p.m. Beijing time on 8/8/2008, the Beijing Olympic Games opened at the National Stadium.

Say what you know about the clock face.

Move the hands and fill in.

The clock face has __ large divisions and __ small divisions.

1 large division for the hour hand is __ hour.

1 small division for the minute hand is __ minute; 1 large one is __ minutes.

When the hour hand moves 1 large division, the minute hand makes exactly __ full turn.

1 hour = __ minutes

Write the times shown on the clock faces.

8 : 08	9 : 00		

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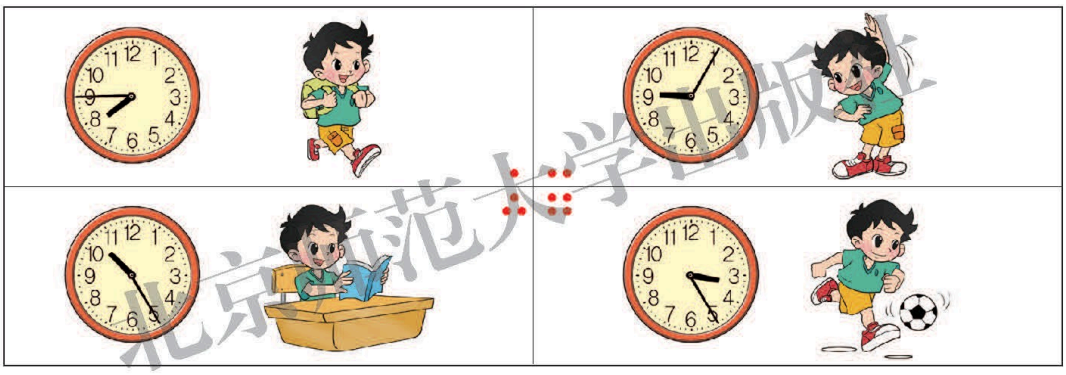
Practice

1. Look and match.

4:12　　9:20　　6:08　　1:30　　8:55

9 : 20	8 : 55	4 : 12	6 : 08

2. Read and say.



3. Calculate and check.

$324 - 218 =$	$126 + 437 =$
$392 + 256 =$	$701 - 264 =$

4. Game.

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How Long Is 1 Minute?

Look and say.

A clock has hour, minute and second hands...

The second hand is the longest and fastest. 1 small division takes 1 second.

How long is 1 minute? Count and fill in.

Tick... tock...

1 second, 2 seconds...

1 min = ____ s

What can you do in 1 minute? Try and fill in.

I clap about ___ times in 1 minute.

I write my name about ___ times in 1 minute.

My heart beats about ___ times in 1 minute.

Count from 1 to 60 and see whose time is closest to 60 seconds.

…, 34, 35, …

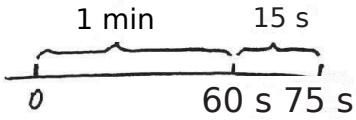
…, 27, 28, …

Try It

In the 10 m air rifle final, each shot must be fired within 75 seconds. How many minutes and seconds is 75 seconds?

1 minute is 60 seconds, and 15 seconds more is 75 seconds.

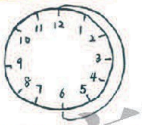
75 s = ___ min ___ s



1 min is 60 s,
75 − 60……

A football match took 1 hour 30 minutes. How many minutes is 1 hour 30 minutes?

One turn is 60 min, and 30 min more is 90 min.



1 h 30 min = ☐ min

1 hour is 60 min…

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Practice

1. Choose and tick (✓).

Eating lunch takes about

20 min ☐
20 s ☐

Brushing teeth takes about

3 min ☐
3 s ☐

Running 2 m takes about

1 min ☐
3 s ☐

2. Try it and fill in.

In 1 min I write about ☐ characters.

In 1 min I walk about ☐ steps.

In 1 min I run about ☐ m.

3. Think and fill in.

1 h = ___ min	1 min = ___ s
90 s = ___ min ___ s	1 min 20 s = ___ s
70 min = ___ h ___ min	1 h 15 min = ___ min

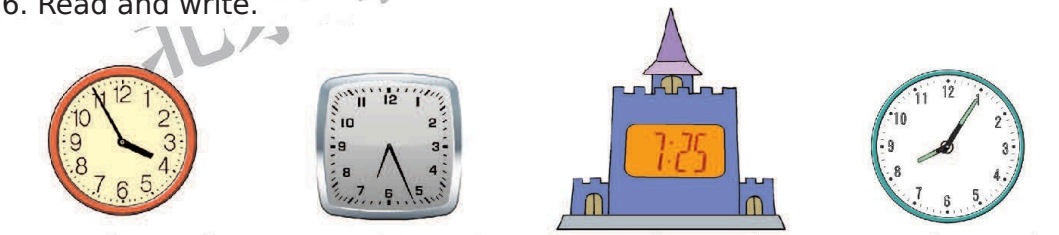
4. Think and fill in.

(1) 1 lesson is ☐ min, break is ☐ min, plus ☐ min makes 1 hour.

(2) The red light stays on for 2 min, then turns green. 2 min is ☐ s.

5. Did they read the times correctly? Talk with a partner.

6. Read and write.



___h ___ min ___h ___ min ___h ___ min ___h ___ min

: : : :

7. Below are the results of four students in the 50 m race.

Xiaoming	Xiaoliang	Xiaoqiang	Xiaogang
8 s	11 s	9 s	12 s

(1) Who ran fastest? Who ran slowest?

(2) How many seconds faster was the fastest than the slowest?

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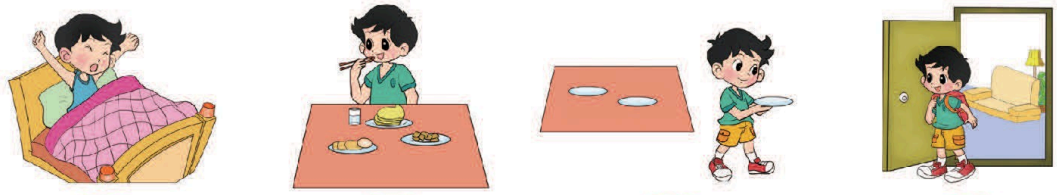
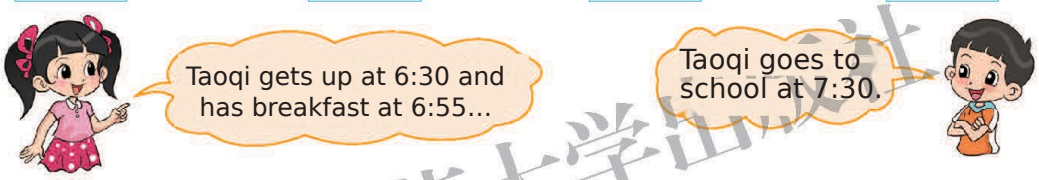
Taoqi's Daily Schedule

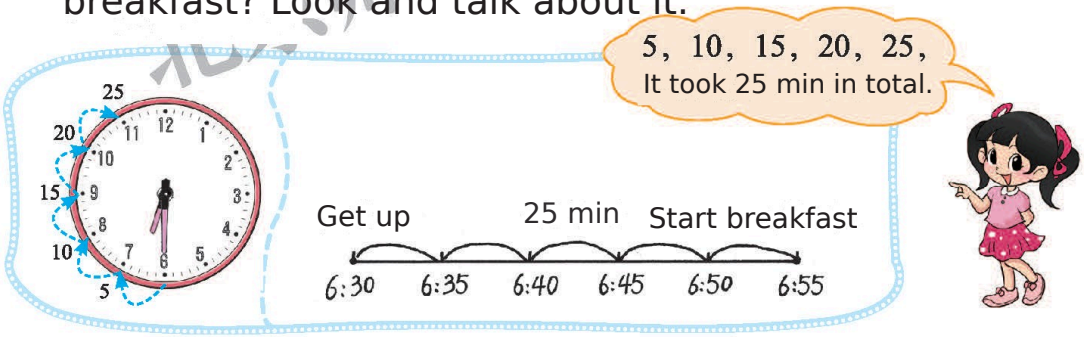
Fill in and talk about it.



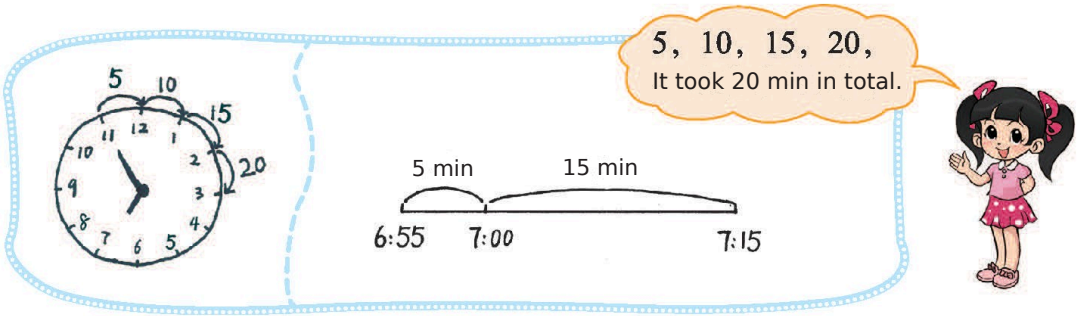
6 : 30 | : | : | :



How long is it from Taoqi getting up to starting breakfast? Look and talk about it.



How long does Taoqi take to eat breakfast? Draw it.



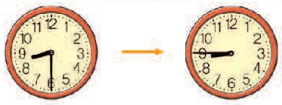
Practice

1. Draw and fill in.

From 9:00 to 11:00,
___ hours passed.

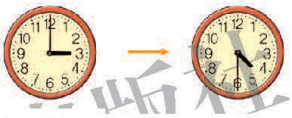
From 10:00 to 10:30,
___ minutes passed.

2. Fill in.



(1) ___ min passed.

(2) What did you do during this time yesterday morning?



___ h ___ min passed.

(2) What did you do during this time yesterday afternoon?

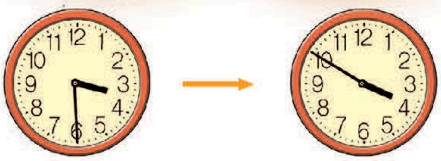
3. Calculate and check.

$318 - 242 =$ $426 + 337 =$

$212 + 356 =$ $720 - 564 =$

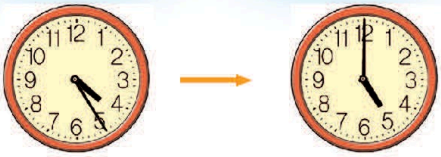
4. After school

(1)



Going home took ____ min.

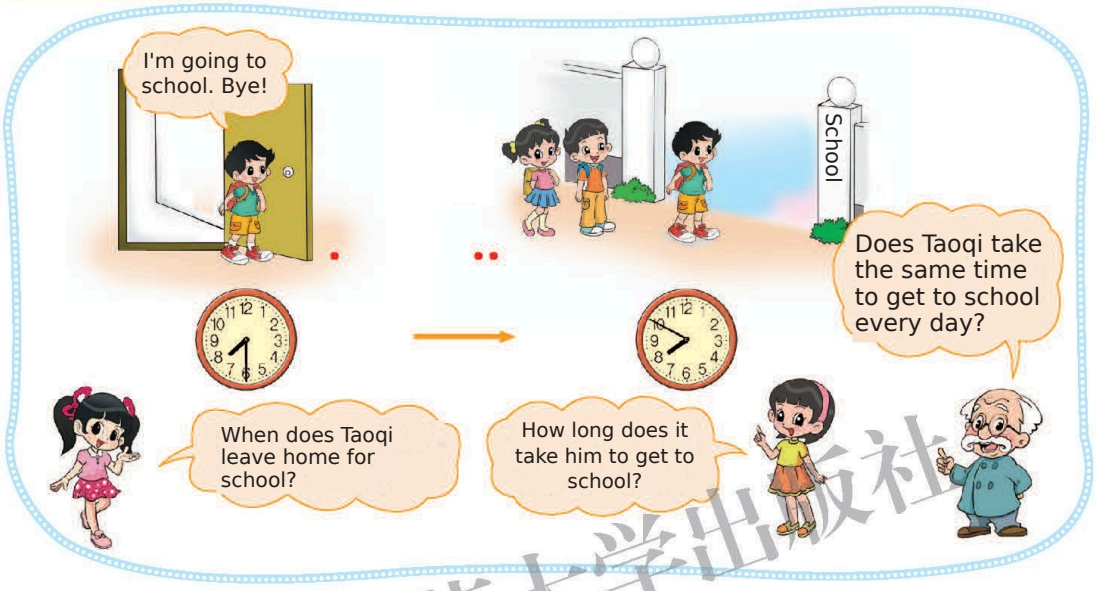
(2)



Played for ___ min.

Maths Is Fun

Time to School

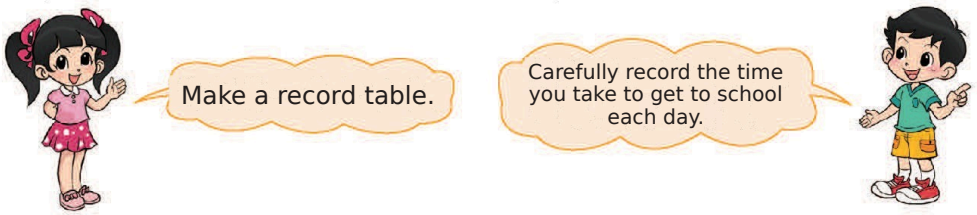


Task

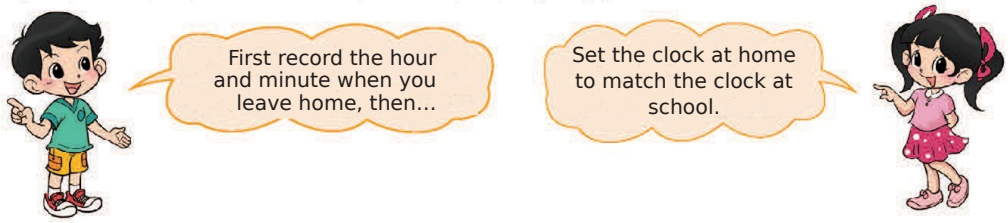
Record the time you take to get to school each day for a week. Is the time the same every day?

Discuss

1. How can we complete this task? Tell your partner what you think.



2. How do we record how long it takes to get to school each day?



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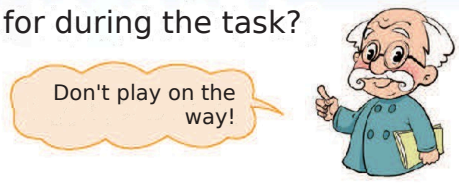
3. What else do you need to watch out for during the task?



Do It

The table on the right shows when Taoqi left home and reached school each day for a week. Help him fill in the time taken.

Time to School Record

	Leave home	Reach school	Time taken
Mon	7 : 30	7 : 53	
Tue	7 : 28	7 : 52	
Wed	7 : 32	7 : 56	
Thu	7 : 30	7 : 55	
Fri	7 : 33	7 : 56	

Think

1. Did Taoqi take the same time to get to school every day this week?

2. What was the longest time Taoqi took to get to school? The shortest? About how long does it usually take?

3. Use the record table in Appendix 3 to record how long it takes you to get to school each day for a week.

4. After a week, look at your group's record tables and talk about how long each of you takes to get to school.

Self-Check

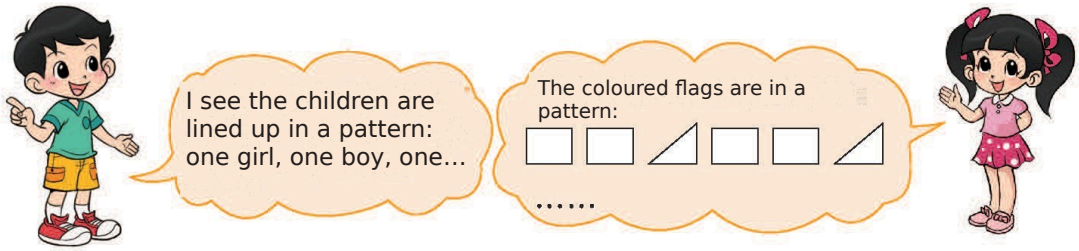
How I did in this activity:

When I meet a problem, I can think about what to do first and what to do next.			
I learned how to measure time and keep a record.			
I can answer questions using the table.			

The Magic of “Repeating”

Showing patterns

1. Look and do. What patterns do you see in the picture? Show them in a way you like.



2. Do you understand the ways of showing patterns below?

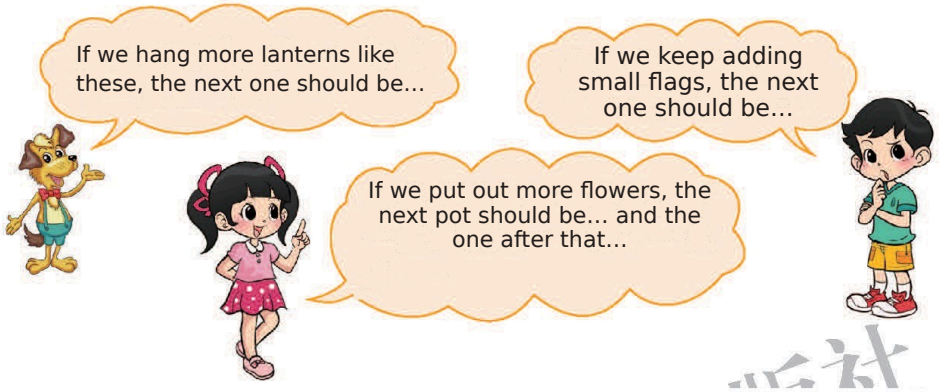
The flower pots are also in a pattern: ✓ ✓ × × ✓ ✓ × ×...

The balloons on the pole are in a pattern. This is how I show it: 112 112 112...

Think and talk about it

1. What do these patterns have in common?

2. Say what comes next in these patterns.



There are many "repeating" things like this in our lives. Can you give some more examples?

Today is Saturday. In 7 days it will be Saturday again.

In music class, we clap the beat: boom cha cha, boom cha cha...

Every year has four seasons.

Our zodiac animals repeat every 12 years.

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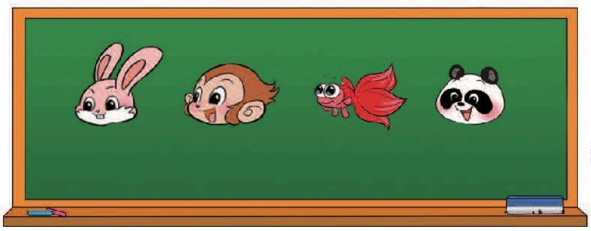
Unit 8 Surveys and Records

Choosing a Mascot

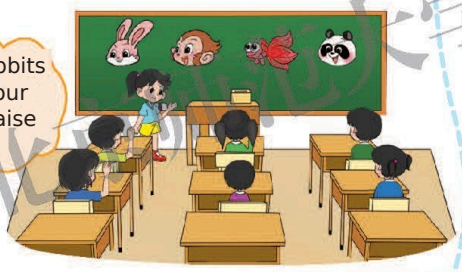


Let's make the animal the class likes best our mascot.

Talk about it: how should we carry out the survey?

To count how many people like each animal best, we can ask everyone to raise their hands...

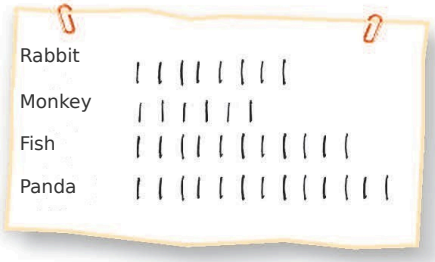
If you like rabbits best, raise your hand. Only raise it once.

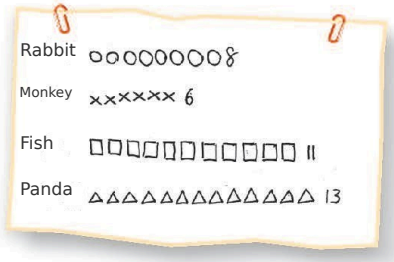


Write down your favourite animal and put it in the box.

Do it Xiaoxiao's way and organise the results.

This is how they recorded the survey data. Do you understand it? Talk with your partner.





Fill in the blanks using your results. Talk with your partner.

( ) people like , ( ) people like monkeys,
( ) people like , ( ) people like pandas.
We can choose ( ) as the mascot.

Practice

1. (1) Tell your partner how you will do the survey.

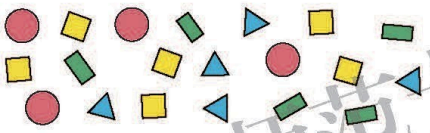
(2) Below is Miaoxiang's survey record.

Comedy	✗✗✗✗✗✗✗✗✗ 9
Magic	○○○○○○○○ 8
Acrobatics	✓✓✓✓✓✓ 6
Dance	△△△△△△△ 7

Miaoxiang surveyed ( ) classmates.

( ) is liked by the most people, and ( ) by the fewest.

2. (1) Count. How many of each are there?



( )	( )	( )	( )

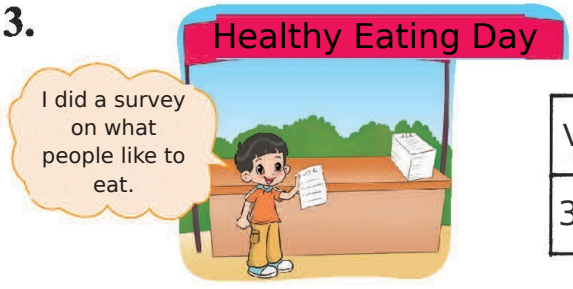
(2) Fill in.

Squares: ( ) more than triangles, ( ) more than circles.

Circles: ( ) fewer than squares, ( ) fewer than triangles.

(3) Most: ( ). Fewest: ( ). The same number: ( ) and ( ).

3.



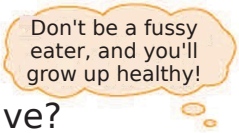
Below is Qisi's survey record.

Veg only	Meat only	Both
3 people	5 people	7 people

(1) Qisi surveyed ( ) classmates.

(2) ( ) has the most, ( ) has the fewest.

(3) Talk about it: what eating advice can you give?



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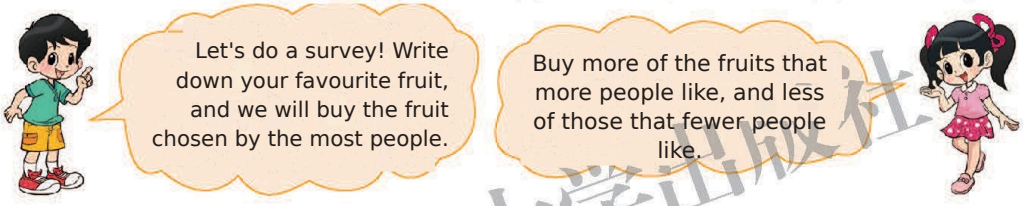
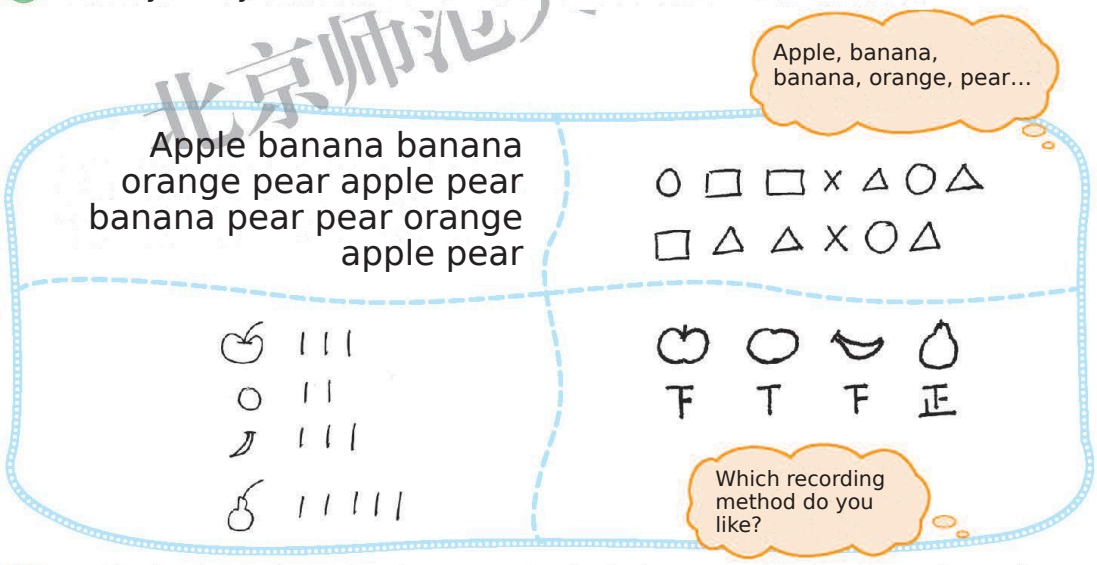
Favourite Fruits

Class 2 (1) will buy apples, oranges, bananas and pears for a party.

How should we buy the fruit? Which should we buy more of, and which less?



Survey everyone's favourite fruit. How can we record the data?



Record everyone's choices in your favourite way. Then fill in.

( ) people like apples, ( ) people like oranges,

( ) people like bananas, ( ) people like pears.

Suggest how to buy the fruit.

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Practice

1. The class is planning a weekend outing.

(1) Talk about it: how should we decide where to go?

(2) Below are Taoqi's and Xiaoxiao's records. Which one do you like?

Zoo	𝍸 𝍸
Museum	𝍸 ||||
Funfair	𝍸 𝍸 ||
Park	𝍸 |||

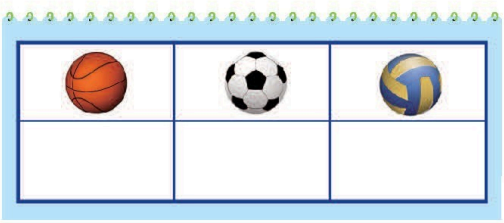
park, funfair, funfair, museum, funfair, park, zoo, zoo, zoo, park, museum, park, park, zoo, park, funfair, funfair, zoo, funfair, park, museum, zoo, museum, funfair, museum, funfair, zoo, funfair, museum, museum, zoo, museum, museum, funfair, zoo, funfair, zoo, funfair, park

(3) Fill in the table below with your survey record and results.

	Recording method	Number of votes
Zoo		
Museum		
Funfair		
Park		

(4) In your survey, most chose ( ), the fewest chose ( ). Suggest where to go.

2.



(1) Record the number of each ball in the table, in your own way.

(2) There are ( ) basketballs, ( ) footballs and ( ) volleyballs.

Final Review

Review and Share

Number and Algebra

On your own

1. Give examples of the "new" numbers you learned about in Year 2.

2. Show the three numbers below in your own way.

Yangtze River: about 6397 km	Yellow River: about 5464 km	Grand Canal: about 1794 km

3. Put the lengths of the Yangtze River, the Yellow River and the Grand Canal in order, and say how you compared them.

4. Think about it: what progress have you made in solving problems?

5. Draw, fill in, and move the beads on the counting frame.

$236 + 148 = \square$

236

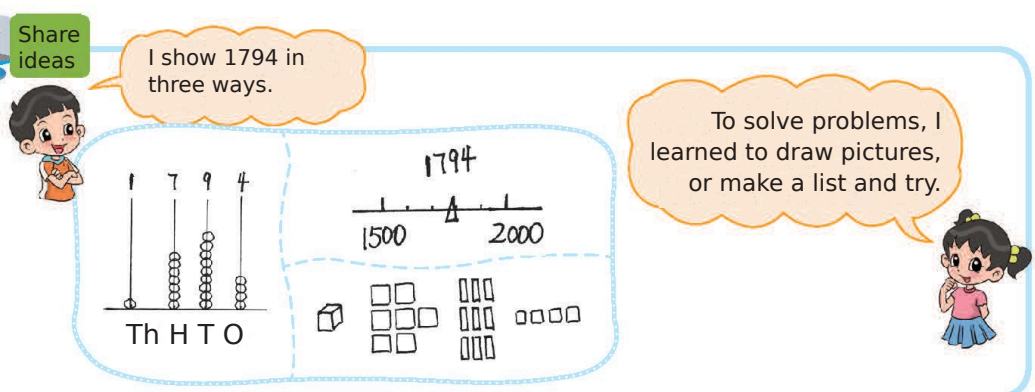
$320 - 157 = \square$

320

6. Sort out the problems you often get wrong. What should you watch out for in division, addition and subtraction? How can you make fewer mistakes?

7. What have you learned about hours, minutes and seconds?

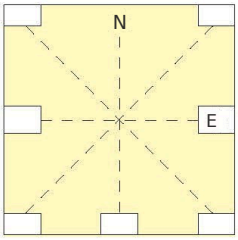
Share ideas

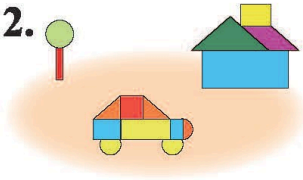


Shapes and Geometry

On your own

1. Talk with your partner about how to tell east, south, west and north in everyday life, and complete the direction board on the right.



2. 

In the picture, trace one of each shape you have learned. Say what features squares and rectangles have, and how you found them.

Share ideas

All four angles of a rectangle are right angles. I checked them with a set square...

All four sides of a square are equal. I found this by folding.

Statistics and Probability

On your own

Survey which of these four ball games your classmates like best.

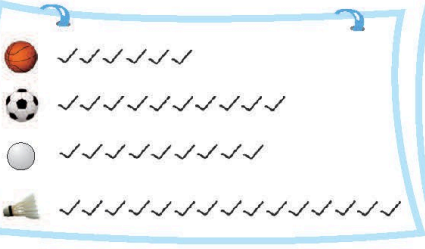
My survey record

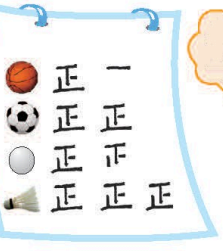
Recording method				
How many like it				

Talk with your classmates. Are their recording methods the same as yours?

Share ideas

I draw ticks ✓





I draw "正".

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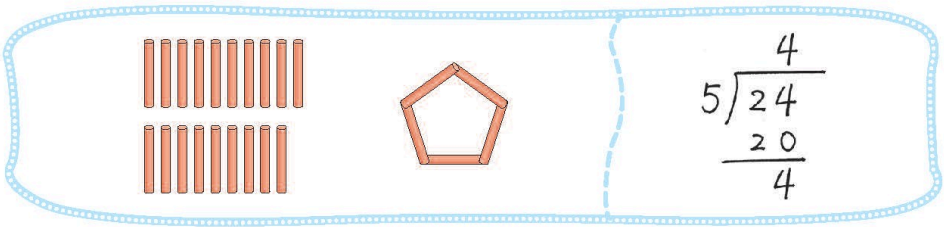
Exercise

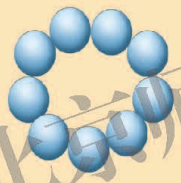
Number and Algebra

1. 5 sticks make 1 pentagon. How many pentagons can 24 sticks make, and how many sticks are left? Think about how you built them, and say what each number in the column method means.



$$\begin{array}{r} 4 \\ 5\overline{)24} \\ 20 \\ \hline 4 \end{array}$$

2. Miaoxiang has 37 beads. At most how many bracelets like this can she make?



3. Put all 41 eggs in egg trays. At least egg trays are needed.

4. Write the underlined numbers in digits and show them on the abacus.

(1) The Oriental Pearl TV Tower in Shanghai is four hundred and sixty-eight metres tall. In digits: ( ) m.

(2) The Beijing–Kowloon Railway is about two thousand five hundred and fifty-three kilometres long. In digits: ( ) km.

(3) The Great Wall runs from Shanhaiguan in the east to Jiayuguan in the west. It is about six thousand seven hundred kilometres long. In digits: ( ) km.

5. Guess: what could the hidden digits be?

32■ < 321　　■827 > 6827　　100■ > 1000

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6. Find the pattern and fill in.

(1) 820, _____, 620, 520, _____。

(2) 225, 250, 275, _____, _____。

(3) 3255, 3155, _____, _____, 2855。

7. Show 3648 in your favourite way.

8. Estimate: about how many words are there? (30, 300 or 3000 words)

Children, did you know? Einstein is a classic example of someone who grew from a "dunce" into a giant of science. As a small child Einstein was known as a "slow child"; at 3 he still could not talk. At 6, when a teacher once called his name, he just stood there dumbstruck, and the class burst into scornful laughter: "What a hopeless dunce." His teacher's report said: "Slow to react, cannot speak clearly, will never amount to anything." In secondary school a teacher concluded: "Whatever he does, he will get nowhere." In his university entrance exam he even failed language and biology, and had to study for another year.

So Einstein became a scientific genius not because he was clever early on, but because he worked hard, loved learning, dared to think for himself and never stopped exploring!

On his 8th birthday his father gave him a compass. The "little dunce" surprised everyone by asking twenty or thirty questions about the compass and its needle. It was this little compass that planted in his heart the seed of science that would later grow into his talent.

9. Use the column method to calculate, then check.

$139 + 258 =$

$400 - 198 =$

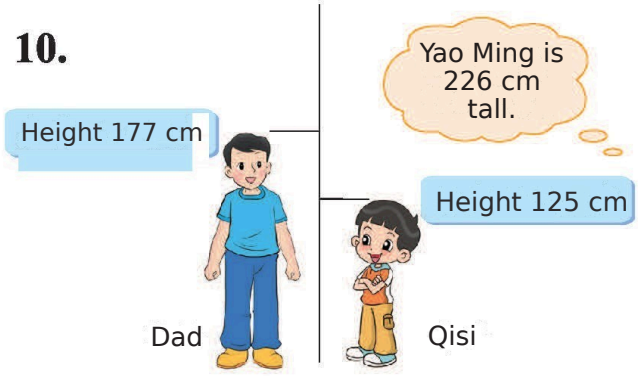
$517 - 486 =$

$372 - 169 =$

$168 + 832 =$

$59 + 745 =$

10.



(1) Mark roughly where Yao Ming's height is.

(2) Ask a maths question and try to answer it.

11. Tick (✓) the best answer.

Eye exercises take about	Getting dressed takes about	Tying laces takes about
5 min ☐	3 min ☐	35 min ☐
5 h ☐	3 s ☐	35 s ☐

12. Visiting an exhibition.

took (20) min spent ( ) min took ( ) min

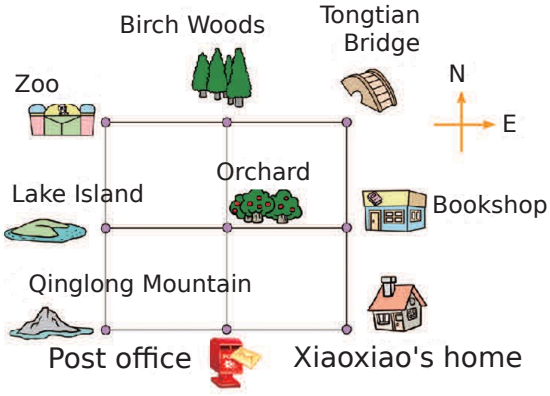
Set off	Arrive	Visit ends	Back to school
7 : 50	:	:	:

Shapes and Geometry

1. Fill in.

(1) The post office is ( ) of the orchard, and the bookshop is ( ) of Xiaoxiao's home.

(2) Xiaoxiao's home is ( ) of the orchard, the zoo is ( ) of the orchard, and Qinglong Mountain is ( ) of the orchard.



2.

7 km = ( ) m	4000 m = ( ) km
2 dm = ( ) cm	30 cm = ( ) dm
60 mm = ( ) cm	5 cm = ( ) mm

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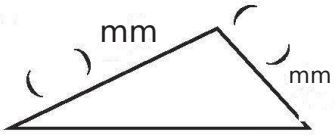
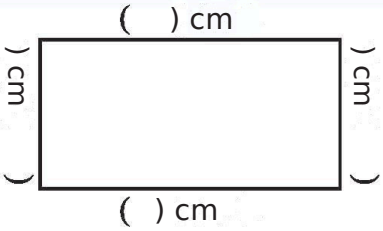
3. Measure.





4. Fill in the appropriate unit in the brackets.

About 30 ( ) tall.

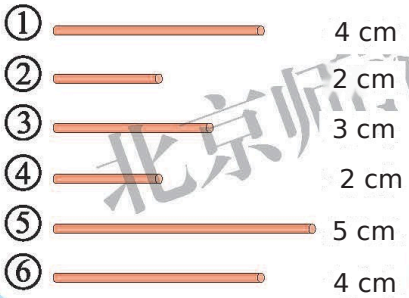
About 60 ( ) an hour.

About 8 ( ) high.

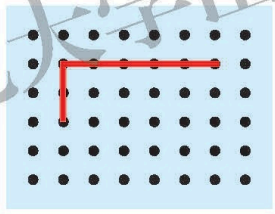
About 3 ( ) thick.

5. Think and discuss.

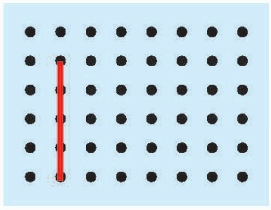
Which sticks can make a rectangle?



Continue drawing on the dot grid.

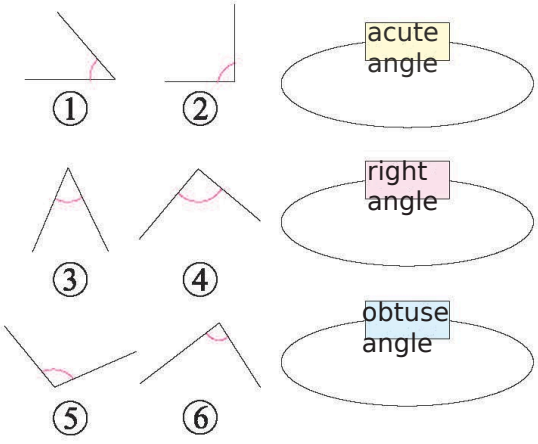


Draw a rectangle.



Draw a square.

6. Write each number in the right circle.



7. Mark three shapes you know in the figure below.

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Statistics and Probability

1. To find out how many pens, pencils, erasers and rulers it sold, Little Bear Stationery made the record table below.

Pens	Pencils	Erasers	Rulers
𝍸 𝍸 𝍸 𝍸 ||	𝍸 𝍸 𝍸 𝍸 𝍸 |	𝍸 𝍸 ||	𝍸 𝍸 𝍸 𝍸 ||||

( ) sold the most, ( ) sold the least. Say how you know.

2. Some older students are collecting data on how fast cars pass their school gate. The table below gives the results for 20 cars.

Car	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Slow		×	×			×			×	×	×		×			×		×		
Medium	×						×	×				×					×		×	
Fast				×	×									×	×					×

How many cars are fast, medium and slow?

3. The animals liked by 5 children are as follows.

Taoqi: rabbit, chicken, bird, tiger and bear;

Xiaoxiao: dog, cat, bear and tiger;

Qisi: dog, tiger, bear and elephant;

Miaoxiang: none;

Lele: cat and bear.

Which animal do the 5 children like most? How do you know? Write down your ideas.

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What did you learn this term?

1. _____

2. _____

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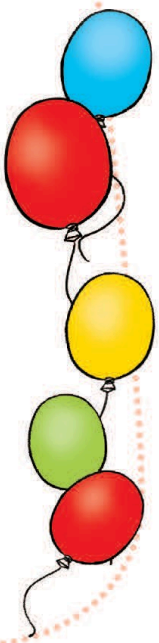
3. _____



Question Bank

What maths problems have you found in daily life and at school? Write them down. Can you solve them?

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Appendix 1

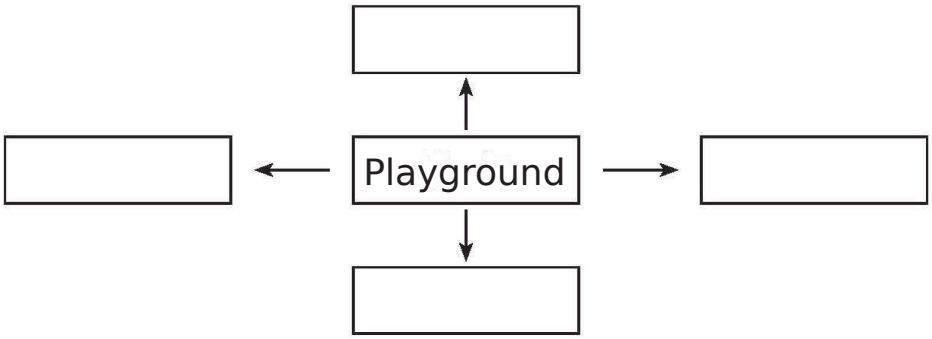

Figure 1

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Figure 2

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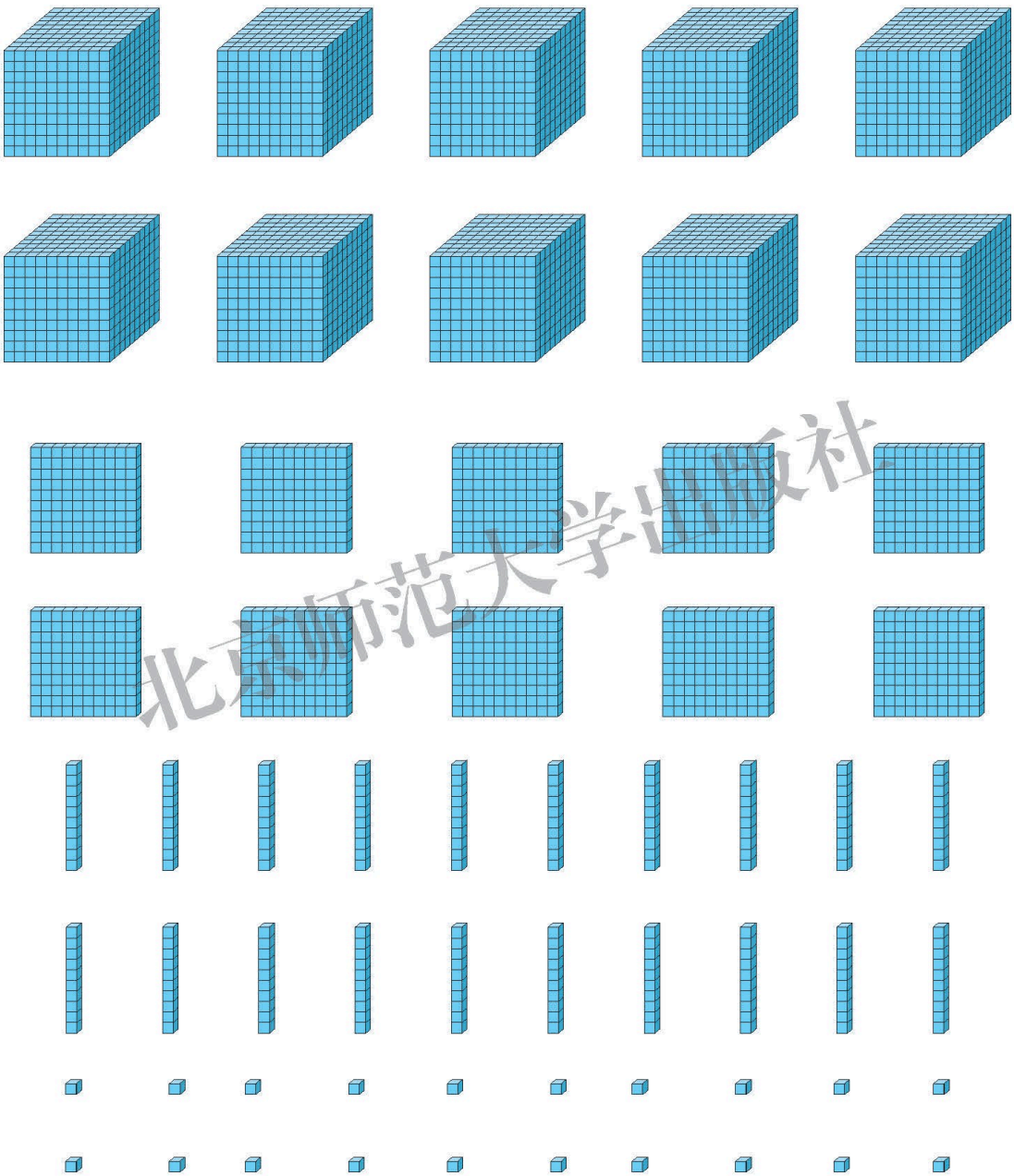
Appendix 2



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Appendix 3

Figure 1

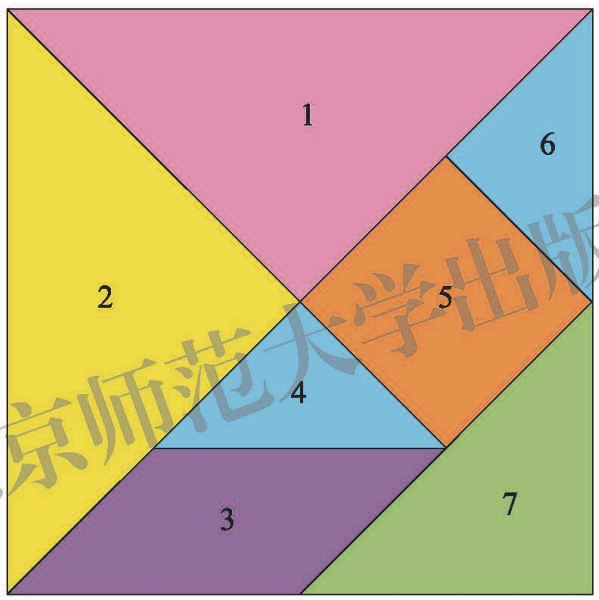

Figure 2

Time to get to school

	Departure time	Arrival time	Journey time
Monday			
Tuesday			
Wednesday			
Thursday			
Friday			

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Afterword

The Compulsory Education Textbooks (Beijing Normal University Edition) were written by many heads and core members of the groups that drew up the national curriculum standards for basic education, together with subject experts, education experts, psychologists and master teachers. They rest on solid research and advanced educational ideas, and offer high-quality writing and professional service. The textbooks strive to reflect the spirit of the national curriculum standards for basic education, make use of many kinds of information resources, show the latest progress in each subject where appropriate, and stress the use of knowledge, skills and ways of thinking in real life. They stay close to students' lives, pay attention to how students learn, meet their different learning needs and promote the all-round development of every student.

The Compulsory Education Textbooks (Beijing Normal University Edition) · Mathematics (Grades 1–6), also known as the New Century Primary School Mathematics Textbooks, fully embody the basic ideas of the new century's reform of the mathematics curriculum, with the healthy growth of children as their fundamental aim. This is the 4th edition of the New Century Primary School Mathematics Textbooks. Building on the strengths of the first three editions, it has been fully revised in accordance with the Mathematics Curriculum Standards for Compulsory Education (2011 Edition).

The new edition has the following distinctive features. Its basic form of presentation is "a situation plus a series of questions", which seeks to make the unfolding of the content, the students' learning, the teachers' teaching and the reaching of the curriculum goals one whole, good for learning and for teaching, so that students keep going through the process of thinking a problem through "from beginning to end". It seeks to give students realistic, interesting and challenging learning materials, to give them time and space to explore and to share their ideas, to show how mathematical knowledge is formed and used, and to meet the needs of different students.

The aim is for students to: appreciate the close links between mathematics, nature and human society; gain the basic knowledge, basic skills, basic ideas and basic experience of activities that they need and that suit their age; develop the ability to find and pose problems and to analyse and solve them; understand the value of mathematics, become more interested in learning it and more confident of learning it well; and gain a first sense of innovation and a scientific attitude.

The chief editors of the New Century Primary School Mathematics Textbooks are Liu Jian, Kong Qiping and Zhang Dan. The other members of the writing group are Hu Guangti, Liu Keqin, Wang Yong, Tao Wenzhong, Wang Mingming, Zhu Dejiang, Chen Xiaomei, Zhang Hong, Qian Shouwang, He Fengbo, Hua Yinglong, Huang Lihua, Zhao Yanhui, Li Tie'an, Wei Huinü, Zhu Yuhong, Ren Jingye and Hou Huiying.

The chief editors of this volume are Wang Mingming and Qian Shouwang. The main people who took part in revising and finalising it were Chen Xiaomei, Hou Huiying, Zhu Dejiang, Huang Lihua and Hu Yu; the final text was compiled and approved by Liu Jian and Zhang Dan. Many local teaching researchers and front-line teachers gave valuable advice on this new round of revision.

Special thanks go to the children Wang Jian'er, Liu Yifei, Chen Taoran, Wang Ziyi, Wu Di and Shang Xinran, and to the teachers Zhu Lei, Guo Chenyue, Xu Yue, Zhu Yufei, Li Lianjie and Zhao Mengjie, who were invited to contribute their work.

We hope that teachers and students will send us their valuable comments as they use the book, so that we can revise and improve it further.

You are welcome to contact us by phone or letter: Primary School Mathematics Editorial Office, Beijing Normal University Press (100088), (010) 58802836, xxsx@bnupg.com; Mathematics Studio, Basic Education Curriculum Research Centre, Beijing Normal University (100082), (010) 58435911, xsjmath@126.com.

Beijing Normal University Press